

1 The multiscale architecture of personal 2 light exposure

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32 Abstract

33 Personal light exposure is associated with sleep and non-communicable diseases,
34 but its architecture remains unresolved. We analysed 191 adults at nine sites in
35 seven countries; 184 contributed 1,478 participant-days, including near-eye data
36 from 141 adults and 816 participant-days and complementary chest data from 154
37 adults and 902 participant-days. Near-eye exposure followed a 24-hour rhythm
38 structured across several scales. The shared local-clock pattern received 78.9% of
39 in-sample model-fit credit, participant patterns 12.9%, participant-day shifts 6.2% and
40 site patterns 2.0%. Separately, participant patterns were 1.80 times as dispersed as
41 site patterns. Reported light source and immediate setting produced large contrasts
42 and received the next-largest allocations after time of day: 24.2% and 30.7% in
43 separate exploratory in-sample decompositions. Site, civil-photoperiod and
44 measured person-level associations were metric-specific. Only 24.0% of waking
45 daytime minutes reached the recommended exposure. This study provides a
46 multiscale observational account of the amplitude and everyday organisation of
47 ocular light exposure.

48 Light reaching the eye supports vision and signals circadian timing, sleep, alertness
49 and neuroendocrine responses¹⁻⁵. Melanopic equivalent daylight illuminance
50 (melanopic EDI) weights illuminance for melanopsin-related sensitivity and is most
51 relevant to these effects^{6,7}. Expert consensus recommends at least 250 lx melanopic
52 EDI at the eye during daytime, no more than 10 lx during the three hours before
53 sleep and no more than 1 lx in the sleep environment for healthy adults with regular

daytime schedules⁸. These physiology-based ranges make daily light a health-relevant exposure, but are not individual risk functions.

Population studies and reviews associate personal light patterns with sleep and major non-communicable diseases⁹. Wrist-sensor studies link light timing and day-night distribution with sleep timing and sleepiness, psychiatric disorders, incident type 2 diabetes, cardiovascular disease and mortality¹⁰⁻¹⁵. Other cohorts link evening, nocturnal or seasonal light with sleep in bipolar disorder and with obesity, diabetes, hypertension and metabolic health¹⁶⁻¹⁹. The exposome framework characterises non-genetic exposures across the life course^{20,21}, yet rarely includes time-resolved ocular melanopic light. Existing studies establish population relevance but mostly measured broad-spectrum light at the wrist or bedside, providing only proxy evidence for melanopic ocular exposure.

An everyday exposure baseline against these recommendations is missing. Personal exposure to natural and electric light follows an expected 24-hour rhythm, including in large wrist-sensor cohorts²², but previous studies focus on particular populations, occupations or settings²³⁻²⁵. Its amplitude and timing across sites and countries, and the contributions of time, site, individuals, day-to-day variation and immediate context, remain uncertain. Daylight availability, satellite estimates and fixed-site measurements cannot resolve these layers as people move through buildings, transport, outdoor spaces and sleep environments²⁶. Sensor position also matters: near-eye sensing is closer to the incident ocular field but is not a retinal measurement; wrist and chest sensors capture related fields shaped by posture, wear and occlusion²⁷⁻³³.

Using the harmonised MeLiDos protocol, we measured personal light across nine sites in seven countries^{34,35}. The primary near-eye sensor characterised ocular light exposure during wear, alongside diaries, light-source and immediate-setting reports, sleep information and questionnaires. The complementary chest sensor provided non-ocular environmental evidence and enabled participation among people reluctant to wear the glasses-mounted sensor. We sought to establish a multisite baseline against the recommendation windows and quantify contributions from local-clock pattern, site and photoperiod, participant and participant-day, immediate environment and routine, and selected person-level correlates. These nested

contributions define the multiscale architecture of personal light exposure. Although not global, the protocol enables harmonised measurement elsewhere, progressively closing gaps across locations, civil photoperiods, latitudes and climates and building towards a global account of ocular light exposure.

Results

A multisite baseline and recommendation adherence

The normalised roster comprised 191 participants across Borås (SE), Delft (NL), Dortmund (DE), Tübingen (DE), Munich (DE), Madrid (ES), Izmir (TR), San José (CR) and Kumasi (GH). Of these, 184 contributed 1,478 participant-days with recorded, non-all-zero light data at either position before quality screening. After screening, the near-eye dataset contained 141 participants and 816 participant-days, the complementary chest dataset 154 and 902, and the common sample 112 participants and 643 participant-days. Near-eye measurements provide primary ocular-exposure evidence during wear, whereas chest measurements are non-ocular; both describe the bedside sleep environment during reported sleep. [Supplementary Table S1](#) gives the sample flow.

The pooled near-eye record followed the expected 24-hour rhythm: low overnight exposure, a morning rise, higher daytime values and an evening decline. Seventeen complementary metrics captured thematic streams rather than one score. Level metrics described daily geometric mean melanopic EDI and the brightest and darkest supported 10-hour means; duration and timing metrics quantified exposure ranges, sustained bright periods, bright or dark windows and threshold crossings. Dose integrated intensity, regularity captured stability and fragmentation, and melanopic daylight efficacy (MDER) described spectral composition relative to visual illuminance. These features are not interchangeable health-response estimates. [Supplementary Table S2](#) and [Supplementary Figure S1](#) give the metric dictionary and site distributions; [Supplementary Figure S2](#) shows a worked derivation. [Figure 1](#) and [Table 1](#) summarise the design and data structure.

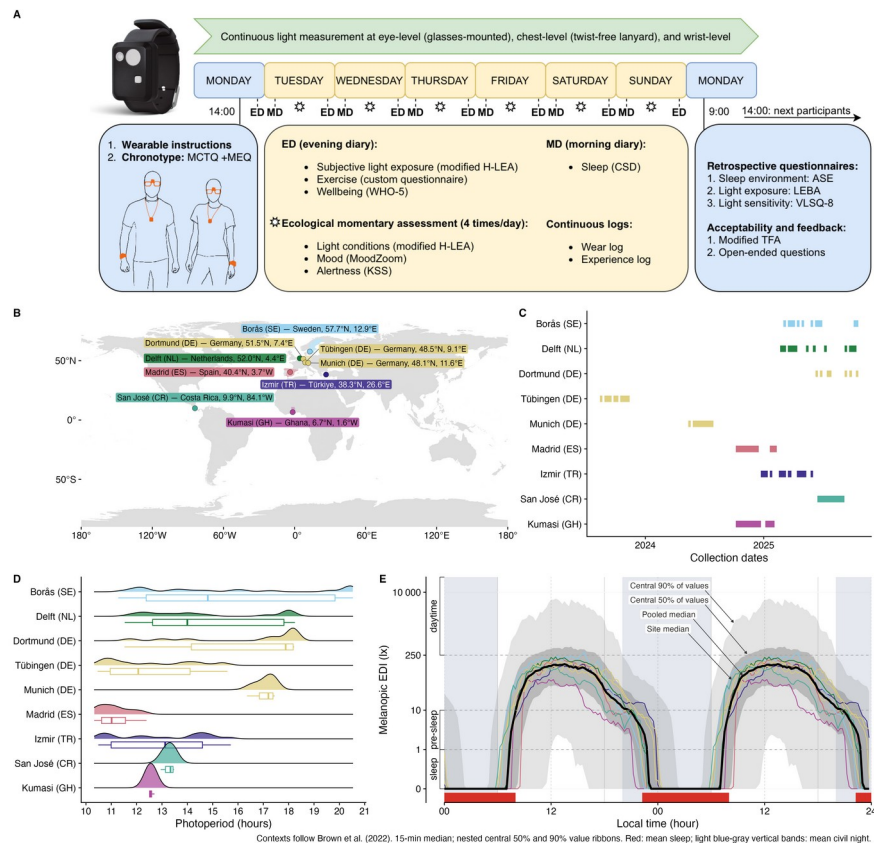


Figure 1: Five-panel overview of the eight-day protocol, sensor positions, nine country-coded sites, collection windows, civil photoperiod, and the pooled near-eye daily profile. A, Eight-day field protocol and sensor positions. The wrist logger is shown as a protocol device but is not an exposure channel analysed in this manuscript. B, Nine sites in seven countries. C, Site collection periods. D, Observed civil photoperiod. E, Site and pooled 15-minute median near-eye melanopic equivalent daylight illuminance profiles from 816 participant-days contributed by 141 participants, repeated over 48 hours for display only. Grey ribbons show the central 50% and 90% of measured values, not confidence intervals; coloured lines are site medians, the black line is the pooled median, and red and blue shading show mean reported sleep and civil night (sun elevation < -6°). During reported sleep, the measurement represents the bedside sleep environment.

128 *Table 1: Participant and measurement characteristics overall and by country-coded study site. The table distinguishes the 191-person roster,*
129 *participants and participant-days with recorded light data, and the screened near-eye and complementary chest samples. Model-specific samples*
130 *are reported with each analysis. Of the 191 roster participants, 184 contributed the 1,478 recorded participant-days. Continuous participant*
131 *characteristics are median with the stated percentile range.*

	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)
Site specifics										
Institution	Overall	RISE	THUAS	BAuA	MPI	TUM	FUSPCEU	IZTECH	UCR	KNUST
Country	—	Sweden	Netherlands	Germany	Germany	Germany	Spain	Türkiye	Costa Rica	Ghana
City	—	Borås	Delft	Dortmund	Tübingen	Munich	Madrid	Izmir	San José	Kumasi
Coordinates	—	57.7°N, 12.9°E	52.0°N, 4.4°E	51.5°N, 7.4°E	48.5°N, 9.1°E	48.1°N, 11.6°E	40.4°N, 3.7°W	38.3°N, 26.6°E	9.9°N, 84.1°W	6.7°N, 1.6°W
Data collection										
Participants ¹	191 roster (g:143 ; c:157; p:116)	17 roster (g:14 ; c:17; p:14)	20 roster (g:13 ; c:15; p:13)	24 roster (g:19 ; c:22; p:19)	26 roster (g:26 ; c:0; p:0)	10 roster (g:10 ; c:10; p:10)	23 roster (g:23 ; c:22; p:22)	17 roster (g:17 ; c:17; p:17)	39 roster (g:6 ; c:39; p:6)	15 roster (g:15 ; c:15; p:15)
Participant-days ¹	1478 roster (g:1134 c:1246; p:902)	137 roster (g:107 c:137; p:107)	125 roster (g:107 c:124; p:106)	176 roster (g:145 c:163; p:132)	208 roster (g:208 c:0; p:0)	80 roster (g:80 c:80; p:80)	182 roster (g:182 c:174; p:174)	138 roster (g:138 c:138; p:138)	312 roster (g:48 c:312; p:48)	120 roster (g:119 c:118; p:117)
Participant time ²	116w 4d	11w 23h	11w 23h	15w 2d	21w 3d	8w 4d	18w 3d	14w 3d	4w 4d	11w 4d
Declared non-wear ²	g:4w 1d (3.66%)	g:2d 14h (3.34%)	g:3d 1h (3.92%)	g:3d 15h (3.42%)	g:6d 10h (4.30%)	g:2d 17h (4.55%)	g:2d 20h (2.20%)	g:3d 3h (3.13%)	g:1d 11h (4.66%)	g:3d 21h (4.80%)
Screened days ^{1,3}	g:816 c:902	g:78 c:96	g:78 c:93	g:107 c:114	g:150 c:0	g:60 c:60	g:129 c:123	g:101 c:102	g:32 c:230	g:81 c:84
Civil photoperiod ^{4,2}	13.25 h (10.62–18.20) (n=1478 d)	14.81 h (11.80–20.49) (n=137 d)	14.00 h (11.88–18.09) (n=125 d)	17.88 h (11.76–18.22) (n=176 d)	12.07 h (10.55–15.41) (n=208 d)	17.20 h (16.45–17.45) (n=80 d)	11.01 h (10.35–12.22) (n=182 d)	13.13 h (10.52–15.57) (n=138 d)	13.34 h (13.01–13.47) (n=312 d)	12.54 h (12.49–12.68) (n=120 d)

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Table continued

	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)
Participant information										
Age ⁴	28 y (21 y–52 y) (n=191 N)	38 y (22.6 y–64.8 y) (n=17 N)	30.5 y (19.9 y–57.1 y) (n=20 N)	35.5 y (20.1 y–58.5 y) (n=24 N)	27 y (22 y–37 y) (n=26 N)	27.5 y (22.2 y–29 y) (n=10 N)	31 y (20.2 y–54.3 y) (n=23 N)	24 y (21 y–29.6 y) (n=17 N)	34 y (22 y–48.1 y) (n=39 N)	23 y (20.1 y–25 y) (n=15 N)
Sex ⁵	F:105 / M:86 (n=191)	F:6 / M:11 (n=17)	F:8 / M:12 (n=20)	F:13 / M:11 (n=24)	F:14 / M:12 (n=26)	F:6 / M:4 (n=10)	F:15 / M:8 (n=23)	F:11 / M:6 (n=17)	F:24 / M:15 (n=39)	F:8 / M:7 (n=15)
Employment status ⁶	F:163 / P:26 / N:2 (n=191)	F:16 / P:0 / N:1 (n=17)	F:11 / P:9 / N:0 (n=20)	F:21 / P:3 / N:0 (n=24)	F:23 / P:3 / N:0 (n=26)	F:9 / P:1 / N:0 (n=10)	F:21 / P:2 / N:0 (n=23)	F:15 / P:2 / N:0 (n=17)	F:34 / P:5 / N:0 (n=39)	F:13 / P:1 / N:1 (n=15)
Chronotype group ⁷	m:62 / i:94 / e:30 (n=186)	m:8 / i:4 / e:5 (n=17)	m:3 / i:9 / e:3 (n=15)	m:8 / i:13 / e:3 (n=24)	m:9 / i:9 / e:8 (n=26)	m:0 / i:8 / e:2 (n=10)	m:5 / i:17 / e:1 (n=23)	m:2 / i:10 / e:5 (n=17)	m:16 / i:20 / e:3 (n=39)	m:11 / i:4 / e:0 (n=15)
Sleep-corrected midsleep on free days ^{4,8}	04:03 (02:08–06:15) (n=185 N)	02:55 (02:04–05:09) (n=17 N)	04:31 (03:05–05:37) (n=15 N)	03:47 (02:17–04:49) (n=24 N)	04:33 (02:41–06:04) (n=26 N)	05:08 (04:12–06:04) (n=10 N)	04:48 (03:10–06:52) (n=22 N)	05:29 (03:35–06:49) (n=17 N)	03:42 (02:17–05:31) (n=39 N)	02:19 (01:22–03:25) (n=15 N)
¹ g, near-eye glasses position; c, complementary chest position; p, paired common sample in which the same participant-day is available at both positions. ² w: weeks; d: days; h: hours; min: minutes. Participant time and declared non-wear use eligible real minutes on screened days; both are shown for near eye only. ³ Screened days are the final participant-days after the 80% completeness screen and exclusion of exact-all-zero melEDI days. ⁴ Median (5th percentile, 95th percentile). ⁵ Sex categories: Female and Male. ⁶ Employment categories: Full/studying, Part/marginal, and Not employed. ⁷ Chronotype groups follow the Morningness–Eveningness Questionnaire score. ⁸ MCTQ: Munich Chronotype Questionnaire; MEQ: Morningness–Eveningness Questionnaire. Clock summaries are circular.										

Recommendation-range coverage differed across windows. Of 573,712 waking daytime minutes, 137,792, or 24.0%, reached at least 250 lx melanopic EDI; of 129,390 minutes in pre-sleep, 81,894, or 63.3%, were at or below 10 lx; and of 383,366 minutes during reported sleep, 336,052, or 87.7%, were at or below 1 lx. The sleep quantity describes the bedside sleep environment because the device was not worn at the eye; the Discussion addresses low-light measurement accuracy. [Supplementary Table S3](#) gives exact numerators, denominators and complementary ranges.

These pooled-minute percentages describe how often exposure lay within range, not participant adherence. Model-based comparisons retained each window's adherent and valid-minute counts.

The main adherence model included 2,216 recommendation-window periods from 140 participants and 782 complete daytime, pre-sleep and sleep cycles, representing 1,028,958 valid minutes. Site-average work-day adherence was 26.5% during daytime, 61.9% during pre-sleep and 90.3% during sleep; free-day adherence was 21.6%, 69.9% and 84.1%. Relative to work days, free-day adherence was 4.92 percentage points lower during daytime (95% confidence interval (95% CI) 2.21 to 7.64 lower), 7.98 higher during pre-sleep (3.28 to 12.67) and 6.13 lower during sleep (3.48 to 8.79). Requiring at least 80% valid coverage preserved each direction and interval conclusion, with a maximum 0.88-point change.

Adherence differences between free and work days varied among sites ($p = 0.003$): during daytime ($p = 0.002$) and sleep ($p = 0.003$), but not pre-sleep ($p = 0.526$). The free-minus-work daytime difference exceeded its site-average value by 15.0 percentage points in Dortmund (DE; 95% CI 6.9 to 23.1; $p = 0.004$) and was 9.2 lower in Madrid (ES; -15.6 to -2.8; $p = 0.042$); the sleep difference was 6.3 higher in Kumasi (GH; 3.4 to 9.1; $p < 0.001$). These directions persisted when the sensitivity analysis required at least 80% valid data coverage.

A descriptive response-scale partition assigned 58.4% to fixed predictors, a further 2.2 percentage points to the participant intercept and 39.4% to observation and distribution variation, giving conditional R^2 of 60.6%. Within fixed-predictor variance, Shapley weights were 94.1% for recommendation window, 4.8% for site and 1.2% for day type. On the design-standardized response scale, site accounted for 2.8

166 percentage points of variance, compared with a 2.2-point participant-intercept
167 increment and 0.7 percentage points for day type. Thus, the site allocation was
168 about 1.3 times the participant-intercept increment and about four times the day-type
169 allocation. [Table 2](#) summarises the main estimates; [Supplementary Figures S4](#) and
170 [S5](#) show site-average and site-specific results.

171 *Table 2: Recommendation adherence by Brown et al. recommendation window and day type. Daytime, Pre-sleep and Sleep identify the*
172 *recommendation windows used for comparison. Daytime comprises diary-defined wake excluding the three hours before Sleep, Pre-sleep*
173 *comprises those three hours, and Sleep uses the reported sleep interval and therefore describes the bedside sleep environment. Pooled minutes*
174 *meeting recommendation report the number and percentage of valid measured minutes whose exposure lay within the applicable range. Work-*
175 *day and free-day percentages, differences, 95% confidence intervals and false-discovery-rate-adjusted p-values are site-average estimates from*
176 *the main recommendation-window model, with each of the nine sites weighted equally. The pooled-minute fractions and modelled*
177 *recommendation-window estimates are distinct estimands. The main model included 2,216 recommendation-window periods from 140*
178 *participants and 782 behavioural cycles, representing 1,028,958 valid minutes. Adherence is defined at the measured-minute or recommendation-*
179 *window-period level and does not classify participants.*

Recommendation adherence by recommendation window and day type

		Observed minute-level adherence	Site-average recommendation-window model			
	Recommendation	Minutes meeting recommendation, n/N (%)	Work day, % (95% CI)	Free day, % (95% CI)	Free minus Work, percentage points (95% CI)	FDR-adjusted p
Daytime	At least 250 lx melanopic EDI during wake, excluding the three hours before sleep	137,792/573,712 (24.0%)	26.5 (24.4 to 28.6)	21.6 (19.3 to 23.9)	-4.9 (-7.6 to -2.2)	<0.001
Pre-sleep	No more than 10 lx melanopic EDI during the three hours before reported sleep	81,894/129,390 (63.3%)	61.9 (58.5 to 65.4)	69.9 (66.3 to 73.5)	+8.0 (+3.3 to +12.7)	<0.001
Sleep	No more than 1 lx melanopic EDI during reported sleep; bedside sleep environment	336,052/383,366 (87.7%)	90.3 (88.7 to 91.8)	84.1 (81.6 to 86.7)	-6.1 (-8.8 to -3.5)	<0.001

Observed pooled-minute fractions and site-average model estimates are distinct estimands. Each of the nine sites contributes equally to the model estimates.

A separate exploratory analysis related adherence across recommendation windows. Within participants, the estimated differences for a daytime-adherence value 10 percentage points above a participant's usual level were -0.26 percentage points for sleep adherence (95% CI -0.95 to 0.43) and -1.09 points for pre-sleep adherence (-2.75 to 0.56). Both intervals included zero, and temporal dependence remained unresolved, so we withheld a within-participant day-level claim rather than interpreting these estimates as evidence of no association. Between participants, each 10 percentage points higher observed average daytime adherence was associated with 2.52 points lower sleep adherence (95% CI 1.48 to 3.55 lower; FDR-adjusted $p < 0.001$) and 3.59 points lower pre-sleep adherence (1.25 to 5.94 lower; FDR-adjusted $p = 0.005$). Directions and interval conclusions remained in the sensitivity analysis, with a maximum 0.60-point change. Thus, participants with higher average daytime adherence tended to have lower pre-sleep and sleep adherence, consistent with higher evening and nighttime light exposure. Complete estimates are reported in [Supplementary Table S4](#), with participant profiles in [Supplementary Figure S6](#).

Personal and day-to-day differences exceed site differences

A nonlinear model decomposed personal light exposure into common local-clock, site and participant curves, and participant-day shifts. Near-eye in-sample R^2 was 0.775. Participant curves were 1.80 times as dispersed as site curves (95% CI 1.16 to 4.33); participant curves plus day shifts had a dispersion ratio of 1.99 (1.29 to 4.77). After accounting for time of day, fitted variation among people and their days within sites exceeded variation among sites.

Conditional in-sample model-fit credit showed the same hierarchy: 78.9% for the shared local-clock curve, 12.9% for participant curves, 6.2% for participant-day shifts, and 2.0% for site curves. Participant curves received 6.41 times the site credit (3.85 to 15.14), and participant curves plus day shifts 9.49 times (5.77 to 21.81).

The shared rhythm did not imply uniform amplitude. Relative to the fitted site-average daily pattern, Borås (SE) ratios were 2.04 to 4.64 from 06:00 to 11:00 and 1.80 to 2.73 from 15:00 to 19:00. Madrid (ES) ratios were 0.15 to 0.47 from 05:30 to

212 10:30 and 0.40 to 0.57 from 17:30 to 20:30. Izmir (TR) shifted from 0.51 to 0.59 at
213 06:00 to 08:30 to 1.98 to 4.28 at 21:00 to 24:00, while Kumasi (GH) was 0.11 to 0.38
214 from 11:30 to 23:00. These are clock-specific departures from the average curve.

215 The chest model had in-sample R^2 of 0.737: common time received 79.1% of model-
216 fit credit, participant curves 11.9%, day shifts 7.0% and site curves 2.1%. Common-
217 sample comparisons supported the combined participant-plus-day ordering, but the
218 participant-only contrast was less stable. Cross-position similarity is complementary
219 context, not replication, equivalence or interchangeability. [Figure 2](#) shows the fitted
220 layers; [Supplementary Tables S5](#) and [S6](#) give both decompositions.

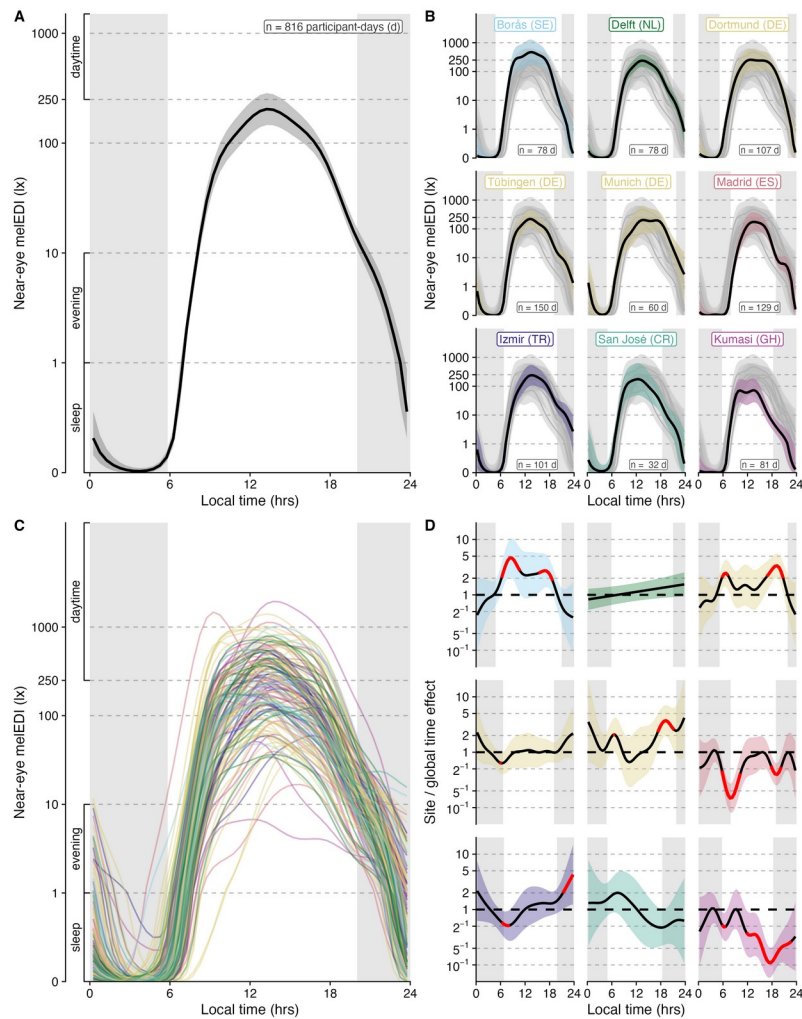


Figure 2: Multiscale daily pattern of near-eye melanopic equivalent daylight illuminance. **A**, Site-average daily curve. **B**, Fitted site curves. **C**, Fitted participant curves after the shared local-clock and site patterns were considered. **D**, Ratio of each fitted site curve to the site-average curve on the melanopic equivalent daylight illuminance plus 0.1 lx scale. Grey bands show mean civil night. Shaded bands are clock-specific 95% confidence intervals; red segments identify displayed half-hours whose interval did not include the site-average curve and are descriptive clock-specific contrasts. The analysis included 141 participants and 816 participant-days across nine sites. The label “global time effect” denotes the site-average local-clock curve (panel **A**).

Geography and photoperiod provide partial context

Site associations were stream-specific. Site differences met the FDR criterion for 8 of 17 primary near-eye metrics: all three level metrics, waking time above 250 lx, the

brightest- and darkest-10-hour midpoints, and the mean and last timing above 250 lx. They were not FDR-retained for dose, spectral composition, stability, fragmentation, sustained bright exposure, the other threshold-duration metrics or first timing above 250 lx. Across metric models, conditional R^2 averaged 36.4%; participant-associated R^2 averaged 25.6% among participant-day metrics. Among the eight metrics with site support, site-associated part R^2 averaged 9.0%. These non-additive components reinforce the stronger individual-level contribution in the daily-pattern analysis.

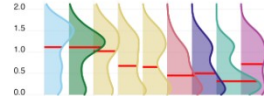
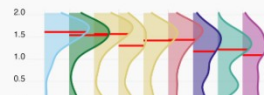
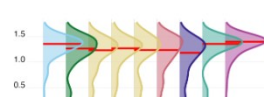
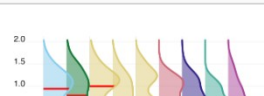
Civil-photoperiod associations were selective. Per additional hour, the fitted daily geometric mean was 1.18 times higher (95% CI 1.11 to 1.26), the brightest-10-hour mean 1.24 times higher (1.13 to 1.36), waking time above 250 lx 1.14 times higher (1.07 to 1.20) and daily dose 1.28 times higher (1.17 to 1.39). Each hour was also associated with 0.134 h less time below 10 lx before sleep (0.066 to 0.201 h less) and the last exposure above 250 lx occurring 0.323 h later (0.186 to 0.460 h). Civil photoperiod met the FDR criterion for 12 of 17 metrics but not stability, fragmentation, sleep time below 1 lx, the brightest-window midpoint or first exposure above 250 lx. Among supported models, photoperiod-associated part R^2 averaged 5.2%; the corresponding mean for the six supported latitude models was 4.0%, meaning that photoperiod had a stronger effect than latitude, when compared across models.

Across the recorded 10.33 to 20.52 h of civil photoperiod, six of nine level, exposure-history, and duration-based metrics changed from increasing to a sustained near-flat tail between 14.24 and 16.20 hours of photoperiod: the daily mean, both 10-hour window means, time above 1,000 lx, waking time above 250 lx and dose. Pre-sleep time below 10 lx, sleep-environment time below 1 lx and the longest period above 250 lx did not. This transition means that the effect of photoperiod becomes uncertain around 14 to 16 hours. The fitted slope became compatible with zero and remained so through the observed maximum. Some classifications were sensitive to site, sample or model form. Several limitations apply as discussed below: latitude was confounded with site, collection periods and photoperiod overlap differed, and weather, built environment, culture and routine could not be separated. The metric-

266 specific results are summarised in [Table 3](#), [Supplementary Figure S7](#) and
267 [Supplementary Tables S7](#) and [S8](#).

268 *Table 3: Near-eye personal light-exposure metrics and their geographic and photoperiod context. Descriptive exposure values are medians with*
269 *interquartile ranges (25th to 75th percentiles). Density plots show site distributions and are descriptive. Metrics are grouped into thematic streams*
270 *and describe distinct features of the daily exposure pattern rather than interchangeable health-response estimates. Model cells report overall-site*
271 *evidence, civil-photoperiod associations with 95% confidence intervals, FDR decisions, participant-associated model information and model-*
272 *specific R^2 summaries where applicable. Site and photoperiod part- R^2 values may contain overlapping fitted information and must not be summed.*
273 *Associations are observational. Absolute latitude is fixed within and confounded with study site, so separate models containing either site or*
274 *latitude are reported; complete latitude and model-adequacy results are provided in the Supplementary Information.*

Near-eye personal light-exposure metrics and their geographic and photoperiod context

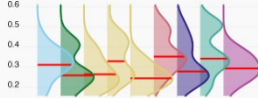
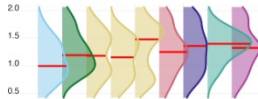
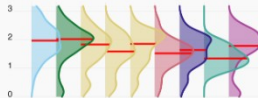
		Descriptive summary		Association evidence		Modelled variation
Definition/relevance		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Duration						
Time above 1,000 lx meEDI	Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	Unit h Median 0.683 IQR 0.183 to 1.558 Participants 141 Days 816		FDR not supported; adjusted p 0.071 Part R-squared 4.7% (2.6% to 12.1%)	Estimate Ratio per 1 h: 1.206 (1.128 to 1.288) FDR supported; adjusted p <0.001 Part R-squared 9.0% (3.7% to 15.7%)	Marginal 21.5% (16.3% to 32.3%) Conditional 50.1% (40.2% to 57.6%) Participant-associated 28.6% (17.7% to 34.2%)
Time above 250 lx meEDI during wake	Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	Unit h Median 2.417 IQR 0.850 to 4.617 Participants 141 Days 737		FDR supported; adjusted p 0.007 Part R-squared 8.6% (5.2% to 18.7%)	Estimate Ratio per 1 h: 1.136 (1.073 to 1.202) FDR supported; adjusted p <0.001 Part R-squared 6.4% (2.0% to 12.7%)	Marginal 18.0% (12.9% to 29.1%) Conditional 50.1% (41.6% to 58.0%) Participant-associated 32.1% (20.7% to 36.7%)
Time below 10 lx meEDI before sleep	Low-light time before bed; limits evening melatonin suppression and circadian delay.	Unit h Median 1.883 IQR 1.025 to 2.583 Participants 139 Days 655		FDR supported; adjusted p 0.049 Part R-squared 5.4% (3.2% to 13.4%)	Estimate Difference per 1 h: -0.134 (-0.201 to -0.066) FDR supported; adjusted p <0.001 Part R-squared 4.9% (1.2% to 10.5%)	Marginal 7.6% (4.6% to 17.2%) Conditional 38.3% (31.8% to 48.5%) Participant-associated 30.7% (21.7% to 38.1%)
Time below 1 lx meEDI during sleep	Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	Unit h Median 7.133 IQR 5.967 to 8.250 Participants 141 Days 778		FDR not supported; adjusted p 0.058 Part R-squared 5.4% (3.2% to 14.2%)	Estimate Ratio per 1 h: 0.987 (0.968 to 1.006) FDR not supported; adjusted p 0.227 Part R-squared 0.6% (0.0% to 3.8%)	Marginal 8.0% (5.1% to 18.2%) Conditional 40.9% (32.8% to 48.5%) Participant-associated 32.9% (22.0% to 37.8%)
Longest period above 250 lx meEDI	Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	Unit h Median 0.633 IQR 0.283 to 1.204 Participants 141 Days 816		FDR not supported; adjusted p 0.398 Part R-squared 2.2% (1.4% to 8.1%)	Estimate Ratio per 1 h: 1.125 (1.065 to 1.187) FDR supported; adjusted p <0.001 Part R-squared 4.9% (1.3% to 9.6%)	Marginal 10.0% (6.1% to 17.9%) Conditional 36.3% (29.9% to 45.3%) Participant-associated 26.3% (18.7% to 33.0%)

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Table continued

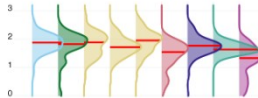
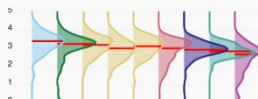
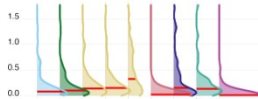
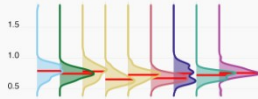
Near-eye personal light-exposure metrics and their geographic and photoperiod context						
Definition/relevance		Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Dynamics						
Interdaily stability	Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	Unit dimensionless Median 0.308 IQR 0.248 to 0.380 Participants 141 Days 816		FDR not supported; adjusted p 0.228 Part R-squared 6.9% (4.1% to 20.4%)	Estimate Odds ratio per 1 h: 0.983 (0.944 to 1.024) FDR not supported; adjusted p 0.401 Part R-squared 0.4% (0.0% to 4.6%)	Marginal 15.4% (10.4% to 30.8%) Conditional 15.4% (10.4% to 30.8%) Participant-associated Not available
Intradaily variability	Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	Unit dimensionless Median 1.253 IQR 0.930 to 1.502 Participants 141 Days 816		FDR not supported; adjusted p 0.483 Part R-squared 5.1% (3.8% to 18.7%)	Estimate Difference per 1 h: –0.019 (–0.056 to 0.019) FDR not supported; adjusted p 0.327 Part R-squared 0.7% (0.0% to 5.7%)	Marginal 5.9% (4.6% to 21.2%) Conditional 5.9% (4.6% to 21.2%) Participant-associated Not available
Exposure history						
meLEDI dose	Intensity–duration-weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	Unit klx·h Median 4.960 IQR 1.936 to 12.313 Participants 141 Days 761		FDR not supported; adjusted p 0.209 Part R-squared 2.8% (1.7% to 8.6%)	Estimate Ratio per 1 h: 1.275 (1.169 to 1.391) FDR supported; adjusted p <0.001 Part R-squared 7.5% (3.1% to 12.9%)	Marginal 11.8% (7.7% to 19.7%) Conditional 33.8% (26.7% to 42.8%) Participant-associated 22.0% (14.5% to 28.7%)

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Table continued

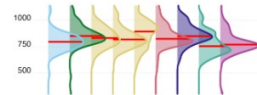
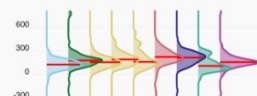
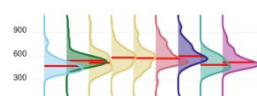
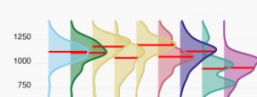
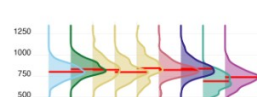
Near-eye personal light-exposure metrics and their geographic and photoperiod context						
		Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Level						
Mean melEDI	Mean melanopic EDI across the day, including zeros; summarises overall exposure while reducing peak influence.	Unit lx Median 5.154 IQR 2.831 to 9.225 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 8.4% (4.8% to 16.6%)	Estimate Ratio per 1 h: 1.181 (1.109 to 1.257) FDR supported; adjusted p <0.001 Part R-squared 7.7% (2.9% to 13.9%)	Marginal 25.4% (18.6% to 34.4%) Conditional 57.7% (51.9% to 65.1%) Participant-associated 32.3% (24.5% to 39.4%)
Brightest 10 h geometric mean	Mean melanopic EDI during the brightest supported 10-hour window; describes the strength of the main sustained daily exposure.	Unit lx Median 110.566 IQR 41.513 to 243.212 Participants 141 Days 816		FDR supported; adjusted p 0.011 Part R-squared 5.7% (3.3% to 12.7%)	Estimate Ratio per 1 h: 1.243 (1.133 to 1.363) FDR supported; adjusted p <0.001 Part R-squared 5.8% (1.9% to 11.1%)	Marginal 17.6% (12.1% to 26.0%) Conditional 46.0% (39.6% to 54.3%) Participant-associated 28.4% (20.9% to 35.4%)
Darkest 10 h geometric mean	Mean melanopic EDI during the darkest supported 10-hour window; describes the sustained low-light period.	Unit lx Median 0.103 IQR 0.020 to 0.253 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 13.7% (9.2% to 23.9%)	Estimate Ratio per 1 h: 1.088 (1.034 to 1.144) FDR supported; adjusted p 0.002 Part R-squared 3.4% (0.5% to 8.3%)	Marginal 22.0% (15.8% to 31.8%) Conditional 61.1% (55.3% to 68.1%) Participant-associated 39.1% (30.4% to 46.2%)
Spectrum						
Melanopic daylight efficacy ratio	Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	Unit dimensionless Median 0.724 IQR 0.643 to 0.795 Participants 137 Days 687		FDR supported; adjusted p <0.001 Part R-squared 9.6% (6.0% to 17.8%)	Estimate Difference per 1 h: 0.024 (0.017 to 0.031) FDR supported; adjusted p <0.001 Part R-squared 13.0% (6.7% to 20.6%)	Marginal 27.8% (21.3% to 38.1%) Conditional 60.5% (54.6% to 68.2%) Participant-associated 32.7% (24.1% to 40.0%)

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Table continued

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
Definition/relevance		Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Timing						
Midpoint of the brightest 10 hours	Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	Unit clock time Median 13:44 IQR 12:48 to 15:00 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 6.5% (4.4% to 12.6%)	Estimate Difference per 1 h: 0.054 (-0.045 to 0.153) FDR not supported; adjusted p 0.305 Part R-squared 0.2% (0.0% to 1.8%)	Marginal 6.7% (4.6% to 13.0%) Conditional 22.9% (16.9% to 31.6%) Participant-associated 16.2% (9.6% to 22.4%)
Midpoint of the darkest 10 hours	Centre time of the darkest 10 hours; indexes the main daily darkness cue.	Unit clock time Median 02:54 IQR 02:01 to 03:50 Participants 141 Days 816		FDR supported; adjusted p 0.020 Part R-squared 4.6% (3.0% to 10.5%)	Estimate Difference per 1 h: -0.148 (-0.245 to -0.050) FDR supported; adjusted p 0.004 Part R-squared 2.1% (0.3% to 5.6%)	Marginal 8.6% (5.9% to 15.8%) Conditional 28.2% (22.1% to 36.7%) Participant-associated 19.6% (12.6% to 25.8%)
First light timing above 250 lx mLEDI	First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	Unit clock time Median 09:08 IQR 08:02 to 10:40 Participants 140 Days 727		FDR not supported; adjusted p 0.112 Part R-squared 3.8% (2.3% to 10.2%)	Estimate Difference per 1 h: -0.097 (-0.247 to 0.053) FDR not supported; adjusted p 0.227 Part R-squared 0.5% (0.0% to 2.8%)	Marginal 6.8% (4.4% to 14.4%) Conditional 31.0% (24.6% to 40.5%) Participant-associated 24.2% (16.2% to 31.5%)
Last light timing above 250 lx mLEDI	Last bright-light exposure; later timing may delay circadian phase and sleep onset.	Unit clock time Median 18:08 IQR 16:27 to 19:42 Participants 141 Days 687		FDR supported; adjusted p <0.001 Part R-squared 11.6% (8.3% to 18.6%)	Estimate Difference per 1 h: 0.323 (0.186 to 0.460) FDR supported; adjusted p <0.001 Part R-squared 4.5% (1.5% to 8.5%)	Marginal 22.2% (17.0% to 30.2%) Conditional 37.9% (31.6% to 46.3%) Participant-associated 15.8% (9.7% to 22.3%)
Mean timing of exposure above 250 lx mLEDI	Average bright-light timing; summarizes the phase of daily circadian stimulation.	Unit clock time Median 13:29 IQR 12:29 to 14:39 Participants 141 Days 742		FDR supported; adjusted p <0.001 Part R-squared 12.7% (9.4% to 19.2%)	Estimate Difference per 1 h: 0.121 (0.028 to 0.213) FDR supported; adjusted p 0.013 Part R-squared 1.2% (0.1% to 3.6%)	Marginal 14.7% (10.7% to 21.9%) Conditional 25.7% (19.8% to 34.2%) Participant-associated 11.0% (5.3% to 17.2%)
Overall distributions are median (interquartile range). Participants is the participant count and Days is the participant-day count. Site and photoperiod part-R ² values can contain overlapping fitted information and must not be summed. Participant-associated R ² is the conditional-minus-marginal contribution associated with the participant random intercept and is undefined for participant-level outcomes. The MDER uses the mean of viable minute-level ratios in both the descriptive summary and geographic-association model cells.						

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279 Immediate environments and routines reorganise exposure

280 Self-reported light source was strongly associated with one-hour near-eye exposure,
281 with an FDR-supported light-source-by-site interaction. Site-average interaction-
282 model estimates were 959.6 lx melanopic EDI for outdoor daylight (95% CI 786.8 to
283 1,170.2), 198.8 lx for indoor daylight (169.3 to 233.4), 85.1 lx for indoor electric light
284 (71.6 to 101.1) and 23.8 lx for a display screen (17.4 to 32.5). Outdoor daylight was
285 11.27 times the indoor-electric estimate (8.71 to 14.60) and exceeded the 250-lx
286 daytime recommendation, whereas the indoor-daylight and indoor-electric intervals
287 were below it. During reported sleep, darkness and external-light estimates were
288 2.24 lx (1.54 to 3.26) and 11.18 lx (8.06 to 15.51), both above the 1-lx bedside upper
289 limit. Outdoor electric light remained imprecise relative to indoor electric light near
290 eye (ratio 1.019, 0.336 to 3.090; FDR-adjusted $p = 0.973$). These contextual
291 averages do not classify people or hours as adherent.

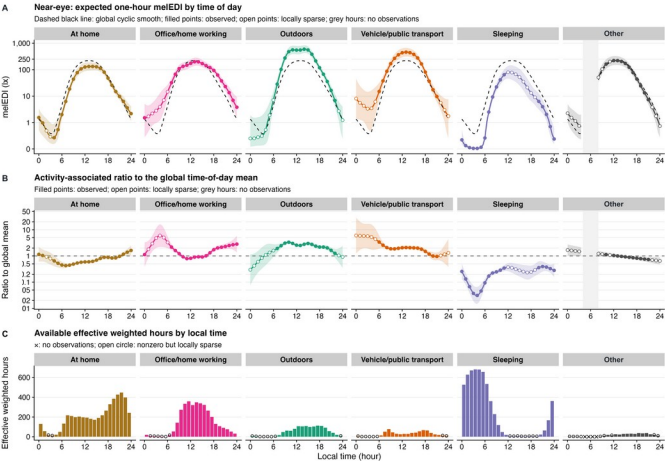
292 External light during sleep in Munich (DE) was estimated at 130.2 lx, 11.65 times the
293 corresponding site average; sleep darkness in Kumasi (GH) was 0.371 lx, 0.166
294 times its site average. Sparse site-category cells limited some estimates.

295 The primary analysis used a population-mean quasi-Tweedie model with participant-
296 cluster-robust covariance for uncertainty; lag-one residual correlation remained
297 0.288. Fitted versus observed exact-zero proportions were 39.8% and 27.8%,
298 respectively. To describe model-based variance allocation, a separate exploratory
299 mixed-effects model added a common participant intercept to site-by-light-source
300 fixed terms. Marginal and conditional R^2 were 0.796 and 0.876, so the intercept
301 added 0.080 on the model-based variance scale. Within marginal R^2 , point
302 allocations were 89.3% to light source, 7.1% to site and 3.7% to their interaction. A
303 Shapley decomposition of an exploratory time-of-day model showed that global time
304 of day received 60.4% and light-source deviations 24.2% of model-fit credit. Light
305 source therefore accounted for 28.6% of the combined credit assigned to these two
306 components. Site received a further 5.2%, participant patterns 7.6%, and day-to-day
307 shifts 2.5%.

308 A separate sensitivity adjusted the population-average light-source comparisons for
309 each participant's overall category distribution, distinguishing hour-to-hour

differences within participants from differences in category composition between participants. The overall category pattern was retained. For external light during sleep, the near-eye ratio relative to indoor electric light changed from 0.449 in the primary additive model to 0.333 (95% CI 0.190 to 0.585), a 25.8% decrease. Thus, between-participant category composition influenced the magnitude, but not the direction, of this contrast. [Supplementary Figure S8](#) shows the temporal, support and site-specific results.

Immediate setting combined behaviour and environment, with an FDR-supported activity-by-site interaction. Site-average near-eye estimates were 714.2 lx outdoors (95% CI 561.7 to 908.2), 331.2 lx on the road or in a vehicle (274.4 to 399.8), 198.9 lx during office or home work (173.9 to 227.5), 76.4 lx while awake at home (62.0 to 94.1) and 4.28 lx during sleep (3.06 to 5.99). The sleep category included many low or zero observations and occasional high stray-light readings, so its fitted mean need not represent a typical observation. Relative to awake time at home, the ratios were 9.35 outdoors, 4.34 for travel, 2.60 for work and 0.056 during sleep. The outdoor estimate exceeded the 250-lx daytime recommendation, the work estimate remained below it, and the sleep estimate exceeded the 1-lx bedside-environment upper limit. Borås (SE) had a fitted outdoor estimate of 3,054 lx, 4.28 times the site average, whereas Tübingen (DE) had 127 lx, 0.18 times the site average. An exploratory mixed model of five setting categories gave model-based marginal and conditional R^2 of 0.766 and 0.858 near eye and 0.768 and 0.859 at the chest; the participant intercept added 0.092 at both positions. Within marginal R^2 , immediate setting received 80.7% near eye and 86.1% at the chest, site 12.5% and 5.9%, and their interaction 6.9% and 8.0%, respectively. [Figure 3](#) shows the immediate-setting patterns and site-specific estimates. In a Shapley decomposition of this near-eye model, time of day received 52.5% and immediate setting 30.7% of model-fit credit. Immediate setting therefore accounted for 36.9% of the combined credit assigned to these two components. Site received a further 5.6%, participant patterns 8.4%, and day-to-day shifts 2.8%.



Bands are conditional posteriorwise 95% intervals; no simultaneous-band or curve-wide inferential claim is made.
Sites are assessed equally on the fitted too-mean scale; participant and participant-day smooths are excluded from displayed curves.
Panel B divides each displayed activity curve by the displayed global time-of-day mean; the dashed reference is 1. Other is display-only.
The activity palette is distinct from the light-source category colours used elsewhere in this report series.
Each multi-select participant-hour contributes 1% to every related activity. Filled circles mark observed support, open circles mark locally sparse support.
Grey regions and x mark zero observations. Only the global smooth is cyclic; thin-plate activity and site deviations may separate over midnight.
No line is wrapped from 24:00 to 00:00. The melEDI axis transformation is display-only.

D Near-eye activity-by-site interaction model

Site-average estimates and site-specific deviation ratios; bold site cells pass the complete near-eye FDR adjustment

	At home P 128 · D 709 · H 5,106	Office/home working P 124 · D 511 · H 3,624	Outdoors P 121 · D 480 · H 1,422	Vehicle/public transport P 111 · D 388 · H 818	Sleeping P 126 · D 714 · H 3,906	Other P 72 · D 158 · H 381
Site-average estimate	76.4 lx (62.6–94.1)	198.9 lx (173.9–227.5)	714.2 lx (561.7–908.2)	331.2 lx (274.6–399.8)	4.3 lx (3.1–6.0)	221.8 lx (97.2–506.4)
Reference category	2.684x home (2.121–3.198)	9.354x home (6.846–12.955)	4.338x home (3.329–5.665)	0.056x home (0.041–0.076)	2.870x home (1.228–7.162)	
FDR p (reference)	FDR p < 0.001	FDR p < 0.001	FDR p < 0.001	FDR p < 0.001	FDR p < 0.001	Additive display only
● Borås (SE)	1.74x (0.94–3.22) FDR p 0.152	1.64x (1.22–2.18) FDR p 0.005	4.28x (2.84–6.45) FDR p < 0.001	1.67x (0.93–3.05) FDR p 0.159	0.45x (0.22–0.96) FDR p 0.084	Not estimable
● Delft (NL)	2.87x (1.38–3.28) FDR p 0.009	1.48x (1.67–2.84) FDR p 0.048	2.32x (1.32–4.09) FDR p 0.014	1.32x (0.96–2.02) FDR p 0.209	1.17x (0.51–2.68) FDR p 0.797	Not estimable
● Dortmund (DE)	2.31x (0.14–4.87) FDR p 0.051	1.67x (0.81–3.96) FDR p 0.169	2.80x (1.35–3.34) FDR p 0.025	2.47x (1.44–4.89) FDR p 0.005	4.53x (1.59–15.79) FDR p 0.048	Not estimable
● Tübingen (DE)	0.42x (0.28–0.63) FDR p < 0.001	0.54x (0.39–0.73) FDR p < 0.001	0.18x (0.11–0.38) FDR p < 0.001	0.34x (0.21–0.56) FDR p < 0.001	0.38x (0.16–0.89) FDR p 0.061	Not estimable
● Munich (DE)	1.25x (0.73–2.14) FDR p 0.486	1.54x (1.00–2.36) FDR p 0.102	1.34x (0.72–2.48) FDR p 0.433	0.81x (0.58–1.13) FDR p 0.299	10.47x (4.40–24.89) FDR p < 0.001	Not estimable
● Madrid (ES)	0.64x (0.46–0.89) FDR p 0.026	1.14x (0.93–1.41) FDR p 0.209	0.97x (0.47–1.98) FDR p 0.953	0.65x (0.33–1.30) FDR p 0.303	0.78x (0.20–3.10) FDR p 0.797	Not estimable
● Izmir (TR)	0.60x (0.56–1.16) FDR p 0.314	0.79x (0.59–1.07) FDR p 0.196	0.97x (0.47–1.97) FDR p 0.953	1.14x (0.96–1.96) FDR p 0.732	1.61x (0.99–2.89) FDR p 0.185	Not estimable
● San José (CR)	0.78x (0.46–1.31) FDR p 0.428	0.50x (0.33–0.76) FDR p 0.006	Not estimable	0.98x (0.53–1.84) FDR p 0.960	0.50x (0.21–1.21) FDR p 0.196	Not estimable
● Kumasi (GH)	0.57x (0.20–1.62) FDR p 0.378	0.67x (0.45–0.99) FDR p 0.085	0.52x (0.33–0.82) FDR p 0.019	0.93x (0.57–1.53) FDR p 0.839	0.16x (0.07–0.42) FDR p 0.001	Not estimable

Site-average estimates and comparisons for the five named categories use the activity-by-site interaction model.
Other is the additive display-only estimate and was excluded from the interaction model.
Site rows are deviation ratios around each category estimate; parentheses show 95% confidence intervals.
Bold site ratios and FDR p-values pass the complete 44-member near-eye family; unsupported cells are not estimable.
Five-category interaction frame: 126 participants; 724 participant-days; 16,135 unique participant-hours; 16,875 generated long rows;
16,135 effective weighted hours; nine sites; 4,748 exact-zero participant-hours. Each multi-select hour contributes 1%.
Other-only display support: P 72 · D 159 · H 391 · P, participants; D, participant-days; H, unique participant-hours.

Figure 3: Personal light exposure by immediate setting across time of day and study sites. The diary variable labelled activity combines behaviour and environmental setting. **A–C**, Exploratory time-of-day analysis. Panel **A** shows site-average expected one-hour melanopic EDI with clock-specific 95% confidence intervals; the dashed black curve is the site-average local-clock smooth. Panel **B** shows each setting curve relative to that smooth, with 1 as the reference. Panel **C** shows effective weighted hours; open circles indicate locally sparse support and grey regions indicate times without observations. **D**, Site-average category estimates and site-specific deviation ratios from the activity-by-site interaction model, with participant-cluster-robust 95% confidence intervals and FDR-adjusted p values. Each multi-select participant-hour contributes weight $1/k$ to every retained setting category. The Other category is sparse and display-only. The fitted frame contains 126 participants, 724 participant-days, 16,135 unique participant-hours, 16,135 effective weighted hours and 4,746 exact-zero participant-hours. Estimates are observational.

Across models of free versus work day, active versus sedentary daily activity status and previous-night sleep duration, the site-average reference was a work, sedentary day following an average 7.8 h of sleep; expected near-eye exposure was 89 lx melanopic EDI (95% CI 67 to 119). Active versus sedentary status was associated with 2.06 times the expected exposure (1.54 to 2.76; FDR-adjusted $p < 0.001$). A model constraining the free-versus-work association across sites estimated 1.45 times higher exposure on free days (1.14 to 1.85; FDR-adjusted $p = 0.004$). Allowing the association to differ by site gave an inconclusive site-average ratio of 1.15 (0.93 to 1.44; FDR-adjusted $p = 0.298$), while some sites differed (interaction $p = 0.017$). Exploratory ratios were FDR-retained in Borås (SE; 1.95, 1.26 to 3.01) and Dortmund (DE; 2.44, 1.42 to 4.22), with FDR-adjusted $p = 0.013$ for each. The free-versus-work result was also inconclusive under the alternative working variance and sensitive to omitting either site. Each additional hour of previous-night sleep duration, centred at 8 h, had an inconclusive ratio of 0.98 (0.85 to 1.12; FDR-adjusted $p = 0.745$). Excluding six participants outside the employment requirement left 15,871 supported hours from 684 participant-days and 131 participants across all nine sites. The free-versus-work, active-versus-sedentary and previous-sleep associations remained stable within model uncertainty; the predictor-by-site interaction for day

type also remained FDR-supported. [Supplementary Table S9](#) and Supplementary Figures [S9](#), [S10](#), [S11](#) and [S12](#) report common, temporal, position-specific and site-specific results.

Person-level correlates are selective

Self-report analyses did not support general exposure profiling. None of 68 prespecified near-eye associations between four light-exposure-related behaviour (LEBA) factors and 17 metrics retained FDR support. Unsatisfactory model checks made the sleep-environment metric unfit for inference, so its estimates were suppressed. Neither average associations nor site interactions across nine visual-light-sensitivity (VLSQ-8) outcomes retained FDR support at either position. Supplementary Tables [S10](#), [S11](#) and [S12](#), and Supplementary Figures [S13](#) and [S14](#), give the full results.

Each hour later in corrected midsleep on free days (MCTQ questionnaire) was associated with first exposure above 250 lx occurring 0.381 h later (95% CI 0.146 to 0.616; FDR-adjusted $p = 0.003$). Each 10-point shift towards greater morning preference (MEQ questionnaire) was separately associated with first exposure above 250 lx occurring 0.444 h earlier (0.189 to 0.699 h earlier; FDR-adjusted $p = 0.001$). The brightest- and darkest-10-hour midpoints followed the same directions; last exposure and the longest-bright-period midpoint did not retain support. The distinct questionnaires were not combined. [Supplementary Table S13](#) and [Supplementary Figure S15](#) show adjusted estimates and observed timing distributions.

Each additional decade of age was associated with higher near-eye brightest-10-hour mean, time above 1,000 lx and daily dose, with ratios from 1.27 to 1.31. These cross-sectional associations may reflect cohort, occupation, behaviour or other confounding, not ageing itself. Neither primary near-eye metric associations with biological sex nor near-eye age-by-site or sex-by-site interactions met the FDR criterion. In a separate daily-pattern analysis, near-eye and chest profiles differed by biological sex (FDR-adjusted $p = 0.028$ and 0.029), but the difference did not meet the FDR criterion after restriction to records with complete immediate-setting data and adjustment. This attenuation is compatible with immediate-setting patterning, but

restriction and adjustment cannot be disentangled as mechanisms. Gender was recorded separately but not analysed; its category matched biological sex in all but one case. Supplementary Tables [S10](#), [S14](#) and [S15](#), and Supplementary Figures [S16](#) and [S17](#), give the complete results.

Measurement position and robustness constrain interpretation

The paired sample of 112 participants and 643 participant-days allowed separate near-eye and chest models to be compared within the same participants and participant-days. The full chest dataset of 154 participants and 902 participant-days broadened coverage beyond the 141 participants and 816 participant-days near eye. At several sites, especially San José (CR), offering a chest sensor enabled recruitment among people reluctant to wear the glasses-mounted sensor. This improved feasibility, not the ocular-exposure sample.

A parallel analysis of overlapping MeLiDos data reported structured position differences across scale, metric, context, site, day and participant³³. Our separate and common-sample analyses likewise show directional agreement but differences in magnitude or support. Chest adherence retained the main daytime and pre-sleep directions and interval conclusions, but provides no sleep ocular claim. We neither pooled positions nor inferred equivalence or a universal correction.

Non-estimable planned quantities were four sleep-environment variants (main plus three sensitivity variants) at each position in the light-behaviour analysis (LEBA), and one longest-period sensitivity comparison incompatible with the metric representation. We retained these outcomes rather than replacing them with zero or null findings. Qualifications for zeros, residuals, dependence, nonlinear-term overlap, sparse support and influence remain with the affected analyses and in Methods.

Discussion

Across nine sites in seven countries, ocular light exposure followed the expected 24-hour rhythm, but its cross-site stability, amplitude and layered sources were unknown. Across analyses, the shared local-clock pattern dominated daily exposure. Separate hourly decompositions showed large light-source and immediate-setting

contrasts, with the next-largest model-fit allocations after time of day. Participant curves and participant-day shifts together exceeded site curves in the daily-pattern model, while site, photoperiod and measured person-level associations were metric-specific. The main contribution of this study is not the rhythm itself, but a multiscale account of its amplitude and of how everyday life reorganises exposure. Because these analyses used different outcomes and decompositions, this is a qualitative synthesis rather than a single quantitative ranking.

The consensus recommendations provide the clearest health-relevant benchmark. Only 24.0% of waking minutes reached the daytime range, while 63.3% of pre-sleep and 87.7% of bedside sleep-environment minutes met their upper limits. Modelled adherence was lower on free days during daytime and sleep but higher during pre-sleep, with site variation in daytime and sleep. Outdoor exposure exceeded the daytime recommendation, but all other conditions remained below it on average. The fitted sleep-category mean exceeded the bedside limit, while 87.7% of pooled bedside sleep-environment minutes met it. These quantities describe different aspects of the exposure distribution and should be interpreted together. These pooled fractions, recommendation-window proportions and hourly means are distinct estimands. Together, they identify intervention opportunities and shortfalls. The recommendations principally address healthy adults aged 18 to 55 with regular daytime schedules⁸.

This baseline connects mechanistic recommendations with population evidence. Large wrist-sensor cohorts have associated day and night light patterns with psychiatric disorders, incident type 2 diabetes, cardiovascular disease and mortality, and light timing with sleep timing and sleepiness¹⁰⁻¹⁵. Other cohorts have linked evening, nocturnal or strongly seasonal light with sleep in bipolar disorder, obesity, diabetes, hypertension and metabolic health¹⁶⁻¹⁹. These outcomes span sleep and major non-communicable diseases. The exposome frames environmental contributions across the life course^{20,21} but rarely includes time-resolved ocular light. This evidence supports personal-light measurement, not clinical risk classification of our exposure categories. Most cohorts used wrist or bedside illuminance without resolving melanopic exposure at the eye, immediate light source or its multiscale architecture.

Exploratory inverse associations of average daytime with sleep and pre-sleep adherence show why the three recommendation windows should not be collapsed into one score. They may reflect schedules, environments or other between-participant differences, not a within-person trade-off or stable phenotype. In exploratory nonlinear models, light source and immediate setting received substantially more in-sample model-fit credit than participant-specific patterns: 24.2% versus 7.6%, and 30.7% versus 8.4%, respectively. This finding is consistent with schedules and immediate environments being prominent correlates of hourly exposure, but it is not a causal explanation for the cross-window adherence association. Longitudinal models of dependence and changing routines are needed to test whether improving one part of the pattern displaces or supports another.

Location mattered, but not as one north-to-south ranking. Site differences concentrated in daily level, waking bright-light duration and several timing metrics, while dose, spectrum and regularity did not show the same site pattern. Civil photoperiod was associated with a broader set of metrics, with larger day-length associations for bright-light level and dose than for the darkest daily window. Between 14.2 and 16.2 h of civil photoperiod, six of nine metrics showed signs of uncertainty about a photoperiod effect beyond that point. Sites were sampled in distinct calendar periods and covered unequal photoperiod ranges, leaving latitude, season, weather, temperature, culture, mobility and the built environment entangled. Repeated seasonal sampling across more sites is needed to separate these influences.

Immediate setting patterned exposure, consistent with environmental opportunity, built settings and behaviour³⁶. Outdoor daylight and being outdoors exceeded indoor electric light and awake time at home, respectively; active days exceeded sedentary days. Site interactions placed outdoors-category exposure higher in Borås (SE) than Tübingen (DE), while the free-versus-work association was most evident in Borås (SE) and Dortmund (DE). Participant intercepts captured remaining between-participant variation, not individual responses or predictions. Future monitoring should preserve local clock time, location, immediate setting, daylight access and site; self-selection, occupation, mobility and weather remain potential confounders.

Light-behaviour and visual-sensitivity scores did not meet their FDR criteria, whereas chronotype aligned with selected timing outcomes and age with selected bright-light and dose metrics. This accords with evidence linking circadian preference or timing to daylight exposure, sleep and routine, without identifying causal direction³⁷⁻⁴⁰. Metric-level biological-sex comparisons were inconclusive near the eye. The complete available-data curve differed by biological sex, but immediate-setting-complete analyses did not retain this separation. Person-level patterns may concern specific exposure dimensions rather than a general high-versus-low phenotype. Near-eye sensing approximates the incident ocular field during wake but is not a retinal measurement; chest sensing samples a related environmental field. Studies show structured differences by body position, morphology, posture and surrounding light²⁹⁻³¹. The matched sample and parallel placement study show metric- and context-specific MeLiDos differences without a universal correction³³. The chest option broadened recruitment among people unwilling to wear glasses-mounted sensors, especially in San José (CR). Study design must balance biological proximity, burden and recruitment while keeping estimands position-specific. The harmonised protocol combined repeated near-eye sensing with local-clock and contextual information across nine sites and seven countries. A generalised additive model separated the shared 24-hour pattern from site, participant and participant-day layers obscured by daily summaries. Collecting these data demanded substantial effort from participants and study teams. Glasses-mounted loggers required fitting, explanation and daily handling; removal, occlusion, charging, discomfort, weather and social acceptability all affected measurement quality⁴¹⁻⁴³. Despite this burden, the protocol measured closer to the incident ocular field at high temporal and contextual resolution. Several limitations qualify these findings. Modest site samples limit site estimates, interactions and subgroup evaluation. Healthy working-age adults may not represent children, adolescents, older or clinical populations, shift workers, people outside employment, or those with restricted mobility or atypical living conditions. Sparse latitude-by-photoperiod coverage leaves latitude, seasonality, culture, climate and built environment entangled and requires broader seasonal sampling. Retrospective hourly setting and light-source reports enabled harmonisation but could not

distinguish fine-grained settings or behaviours within home, work or transport. Very low readings require caution. The manufacturer specifies an operating range starting at 1 lx, and independent laboratory testing began at 2 lx⁴⁴. A scene-based field validation found high overall agreement with laboratory-grade spectral references, but underestimation increased at lower intensities and varied with lighting, time, scene complexity and site⁴⁵. Single-digit values therefore approximate low-light exposure rather than provide precise measurements. Finally, no health outcome was measured.

This nine-site reference locates near-eye exposure relative to physiology-based recommendations, shows that variation among people and days within sites matters, and identifies temporal, contextual and positional information for future cohorts. Taken together, everyday ocular light exposure is structured, measurable and strongly shaped by behaviour and micro-environment, but cannot be reduced to site, latitude or photoperiod. Next steps are broader sampling across photoperiods, latitudes and locations and prospective links to sleep, mental, metabolic, cardiovascular and other outcomes. Causal health benefits or policy effects require interventions and representative longitudinal designs. Potential interventions through clinical advice, occupational health, building design, urban planning or public-health guidance must address the contexts in which people receive light. Time-resolved near-eye measurement therefore offers a practical route from circadian biology to precision prevention and context-sensitive environmental health.

Methods

Study design and sites

MeLiDos is a prospective multicentre observational field study of personal light exposure conducted with a harmonised protocol³⁵. Participants completed an eight-day ambulatory protocol comprising continuous light logging, repeated smartphone assessments and daily diaries under free-living conditions. At an in-person registration visit, participants provided consent, completed baseline questionnaires, received study instructions and were fitted with the loggers. Seven consecutive recording days separated registration and return visits; most sites recorded from Monday to Monday, while Dortmund (DE) recorded from Tuesday to Tuesday. If a

return visit was delayed, participants continued until formal study termination and device retrieval. Data were collected between August 2023 and October 2025 at Borås (SE), Delft (NL), Dortmund (DE), Tübingen (DE), Munich (DE), Madrid (ES), Izmir (TR), San José (CR) and Kumasi (GH). Sites were an opportunity sample of consortium partners and were not selected to represent national populations. Questionnaire materials were administered in the local language except in Tübingen (DE) and Kumasi (GH), where English was used, and translated materials followed a harmonised translation, review, adjudication, pretesting and documentation process⁴⁶.

Participants, recruitment and ethics

The registered target population comprised adults aged 18 to 65 years in full-time employment or part-time employment above 80%. Exclusion criteria included diagnosed psychiatric or sleep disorders, shift work in the preceding two months, regular tobacco or recreational drug use, photosensitising medication, visual impairment incompatible with the devices and residence outside the local study area during recording. The retained near-eye sample included nine participants outside the registered age or employment criteria: one participant older than 65 years, two recorded as not employed and seven recorded as marginally employed; one participant met both the age and employment criteria for exclusion. As a near-eye-only eligibility sensitivity, we refitted the additive and predictor-by-site models after excluding participants recorded as not employed or marginally employed. Students and trainees were retained, and age above 65 years was not an exclusion criterion for this sensitivity. The target was at least 15 participants per site, based on prior power calculations and allowing for up to one-third dropout or incomplete recording⁴⁷. Participants were recruited through local advertisements and institutional mailing lists. At some sites, especially San José (CR), reluctance to wear the glasses-mounted near-eye sensor limited recruitment, and the chest option broadened recruitment possibilities.

All participants provided written informed consent. Approvals were obtained from the Ethics Committee of the Technical University of Munich for Tübingen (DE; 2023-115-S-KK) and for Munich (DE) and Dortmund (DE; 2024-118-S-SB); the Dortmund (DE)

site at the Federal Institute for Occupational Safety and Health (BAuA) operated under this TUM multicentre approval. The other approvals were from the Swedish Ethical Review Authority for Borås (SE; 2024-06792-01), the THUAS Ethics Committee for Delft (NL; positive review dated 29 May 2024), the Research Ethics Committee of Universidad San Pablo-CEU for Madrid (ES; 834/24/106), the Scientific Research and Publication Ethics Committee of the Faculty of Science and Engineering at İzmir Institute of Technology for Izmir (TR; decision 5/1), the Comité Ético Científico of Universidad de Costa Rica for San José (CR; CEC-279-2025), and the Committee on Human Research, Publications and Ethics of Kwame Nkrumah University of Science and Technology for Kumasi (GH; CHRPE/AP/644/24). The protocol was first trialled with participants at the Tübingen (DE) site, whose feedback informed procedural refinements before implementation at the remaining sites; further details are reported in the published protocol³⁵.

Personal light measurements

Personal light was recorded every 10 s with spectrally sensitive ActLumus loggers (Condor Instruments, São Paulo, Brazil). Eight visible-band channels were processed by the device to generate photopic and alpha-opic quantities, including melanopic EDI⁶. The manufacturer specified a factory-calibrated operating range of 1 to 100,000 lx and 10% precision at 1,000 lx. Independent testing against a criterion spectrometer across indoor LED and outdoor daylight conditions covered 2 to 100,000 lx and reported mean photopic error of +3.3% (SD 13.4%) for ActLumus⁴⁴. A subsequent scene-based field validation found high overall agreement with laboratory-grade spectral reference instruments, together with systematic underestimation that varied with intensity and setting⁴⁵. Participants wore the primary device centrally on non-prescription spectacle frames near the eye and, where available, a complementary pendant-mounted device at the chest. The near-eye position measures light closer to the eye but is not a direct retinal measurement. The chest position provided a separate measurement of the local environmental light field. Paired common-sample analyses evaluated how closely it tracked the primary near-eye measurement as a potential ocular proxy. During reported sleep, devices were placed facing upward on a bedside surface near head level, so both positions

characterise the bedside sleep environment. Participants recorded removals and placed devices in an opaque bag during waking non-wear^{43,48}.

The placement analysis from the same project used overlapping MeLiDos data and is cited as related methodological evidence, not independent replication³³. No analysis pooled sensor positions or estimated a universal conversion. Common-sample analyses retained the same participants and participant-days or participant-hours at both positions and fitted each position separately.

Contextual and person-level measures

Participants completed the core Consensus Sleep Diary after awakening⁴⁹, together with evening reports and paper diaries recording light source and an activity-environment category for each hour. Smartphone assessments of sleepiness and mood were scheduled at 11:00, 14:00, 17:00 and 20:00, and evening questionnaires assessed wellbeing and physical activity^{50–52}. Smartphone measures and baseline or end-of-study questionnaires used MyCap integrated with REDCap^{53,54}. Participants photographed and uploaded the paper hourly diaries each day.

The modified Harvard Light Exposure Assessment categories were indoor electric light, outdoor electric light, indoor daylight, outdoor daylight including shade, an emissive display, darkness during sleep and external light during sleep⁵⁵. The primary light-source analysis used one retained category per participant-hour. The diary variable labelled activity combined behaviour and environmental setting. Its categories distinguished sleep, awake time at home, motorised travel, active travel, office or home work, outdoor work, outdoor leisure and other activities. We therefore refer to this variable as immediate setting in the narrative. For analysis, outdoor travel, outdoor work and outdoor leisure were combined as Outdoors. Reports could contain concurrent categories; duplicates were removed and an hour with k retained labels contributed weight $1/k$ to each, preserving total weight one. Day type was coded as work or free day, daily activity status as sedentary or active, and previous sleep duration was linked from the preceding sleep period.

Baseline measures recorded age, biological sex, gender, employment and chronotype; end-of-study measures covered visual light sensitivity, the sleep environment and habitual light-exposure behaviour. Chronotype was represented

separately by corrected midsleep on free days from the Munich Chronotype Questionnaire and the Morningness-Eveningness Questionnaire, whose higher scores indicate greater morning preference^{56,57}. Visual sensitivity used the VLSQ-8⁵⁸, sleep environment used the Assessment of Sleep Environment⁵⁹, and four Light Exposure Behaviour Assessment (LEBA) factors represented outdoor time, device use in bed, ambient light before bed and morning or daytime light use⁶⁰. Biological sex and gender were recorded as separate variables. Models analysed biological sex, coded Female or Male, but did not analyse gender; its recorded category matched biological sex in all but one case.

Time, sleep/wake status and coverage preparation

Processing preserved both actual time in Coordinated Universal Time and local wall-clock time. Actual time ordered observations and measured elapsed gaps; local time aligned daily patterns across sites. Repeated local minutes during autumn clock changes remained distinct in actual time, and structurally absent spring minutes remained missing. Melanopic EDI values at or above the validated 100,000-lx operating boundary were unavailable analytically but retained in the raw channel with a flag.

Diary-defined sleep began at reported sleep preparation and ended at reported wake. Sleep values were retained as the intended bedside record. Declared waking non-wear made the analytical light value unavailable; missing wear labels were not treated as non-wear. Complete 1,440-minute local participant-day grids preserved unsupported periods as missing rather than deleting rows or replacing them with zero. A day was eligible when at least 80% of this combined waking and bedside-sleep record had valid melanopic EDI. Hourly summaries required at least 50% valid minutes in that hour, but an unsupported hour did not erase valid minutes from the daily decision or minute-based metrics. Two otherwise eligible near-eye days and three chest days with finite but entirely zero melanopic EDI were excluded by the defined exact-all-zero screen.

681 Exposure metrics

682 Metrics were derived in R⁶¹ with LightLogR^{62,63}. The 17 outcomes represented
683 complementary properties of the daily pattern. Three level metrics described the
684 daily geometric mean melanopic EDI and the mean melanopic EDI in the brightest
685 and darkest 10-hour windows. Five duration metrics quantified time above 1,000 lx,
686 waking time above 250 lx, pre-sleep time below 10 lx, sleep-environment time below
687 1 lx and the longest continuous period above 250 lx. Five timing metrics located the
688 brightest and darkest windows and the first, last and mean timing above 250 lx.
689 Melanopic EDI dose integrated exposure over time, melanopic daylight efficacy ratio
690 characterised spectral composition relative to visual illuminance, and interdaily
691 stability and intradaily variability characterised regularity and fragmentation.
692 Geometric means were calculated from $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$, followed by
693 inverse transformation and removal of the same offset⁶⁴. All-observed-zero windows
694 were retained as exact zero within verified numerical tolerance. Metric-specific
695 support rules made only the affected metric unavailable, not the participant-day. A
696 longest-period duration remained a lower bound when missing data could conceal a
697 longer period; exactly identified periods were examined separately.

698 The melanopic daylight efficacy ratio was the arithmetic mean of viable one-minute
699 melanopic EDI to photopic-illuminance ratios on a complete 1,440-position local-
700 clock grid. Both channels had to be finite and strictly positive, and at least 720
701 positions were required. Repeated autumn minutes were averaged within each
702 channel before division, and absent spring minutes remained missing.

703 Recommendation-window and adherence analyses

704 We compared valid one-minute values with the Brown et al. recommendation
705 ranges⁸: at least 250 lx melanopic EDI during daytime, no more than 10 lx during
706 pre-sleep and no more than 1 lx during sleep. Daytime comprised diary-defined
707 wake excluding the three hours before reported sleep, pre-sleep comprised those
708 three hours and sleep comprised the reported sleep interval. Each descriptive
709 numerator was divided by all valid primary-record minutes in the corresponding
710 window and pooled across the analysed sample. Daytime and pre-sleep used the
711 worn near-eye record; sleep used the bedside sleep-environment record.

The main adherence analysis retained one row per window period and defined adherence as adherent valid minutes divided by all valid minutes in that period. A behavioural cycle crossed midnight when necessary and inherited the work or free label of the calendar date on which daytime began. The any-valid primary sample retained a window with at least one valid comparison; an otherwise identical coverage sensitivity required at least 80% coverage of the exact window interval. Exact 0% and 100% periods remained in an endpoint-inflated beta-binomial model. Fixed predictors represented the complete window-by-site-by-day-type pattern, and the retained random effect was a participant intercept in the beta-binomial mean. Population averages weighted all nine sites equally. Separate FDR families covered window-specific free-minus-work contrasts, interaction tests, site-specific contrasts against zero and 27 comparisons of site-specific contrasts with their window-specific equal-site values. Chest models were fitted separately for daytime and pre-sleep.

The exploratory cross-window extension used daytime adherence as the predictor and sleep or pre-sleep adherence as outcomes in daytime-anchored behavioural cycles. It separated each participant's cycle-specific daytime deviation from their observed average daytime adherence, while adjusting for window, site, day type, their corresponding interactions and the participant's mix of daytime observations across day types. The four within-participant and between-participant associations formed one FDR family. Positive serial dependence remained unresolved and some unchanged sensitivity fits were non-estimable, so no within-participant day-level claim was stated. Descriptive profiles used site-average participant means for 139 complete three-window profiles and 417 participant-window points; they neither identify nor rank participants and do not define stable participant types.

Daily temporal architecture

The daily-pattern analysis used supported 30-minute arithmetic means, retaining a bin when at least 15 one-minute values were valid. The outcome was $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$. A generalised additive model fitted with `mgcv::bam()` and `fREML` combined a cyclic common local-clock smooth, sum-to-zero site curves, participant curves and participant-day random intercepts^{65–68}. A boundary-aware first-order autoregressive working structure prevented residual sequences from crossing participant-day, clock-repeat or gap boundaries. Fitted-curve dispersion was

compared on a common 48-bin grid. Conditional Shapley allocation averaged each component's in-sample model-fit credit across all component orders, thereby distributing shared fitted information rather than assigning causal or predictive importance^{69,70}. Confidence intervals for dispersion and allocation ratios used 2,000 hierarchical cluster resamples while holding the fitted model structure fixed.

Geographic and photoperiod analyses

We analysed 17 participant-level or participant-day metrics with metric-specific Gaussian or Tweedie exponential-dispersion models^{71,72}. We treated overall site differences, civil photoperiod, absolute latitude and the adequacy of a linear latitude term relative to site as four separate model questions, each adjusted as a complete 17-test FDR family. Participant-day models included participant random intercepts. Mixed-effects models were fitted with `lme4` or `glmmTMB`, and component R^2 values were obtained with `performance` where reported^{73–75}. Site and latitude were not entered as independent fixed predictors in one model because each site has one latitude; site and latitude models instead used the same observations and photoperiod adjustment. Site follow-up contrasts were calculated with `emmeans` only after a supported overall site test and compared each site with a site-average estimate, obtained by giving each included site equal weight⁷⁶.

Nonlinear photoperiod analyses fitted each of nine planned metrics separately with a generalised additive model estimated by restricted maximum likelihood, a thin-plate photoperiod smooth with basis dimension six and participant random effects nested within site. The derivative, meaning the slope of the fitted curve, was evaluated with `gratia` at 100 equally spaced photoperiod values with 95% intervals⁷⁷. A qualifying descriptive transition required the preceding slope interval to be wholly above zero, the next interval to include zero and every later interval through the recorded maximum to include zero as well. This procedure located a change in the fitted slope but did not test equivalence, a mechanism or a physiological ceiling. Expanded-basis, fixed-site, common-sample and leave-one-site-out analyses assessed stability. The gap-timing-unaware sensitivity dataset still passed the 50%-per-hour and 80%-per-day rules, but did not use the timing of remaining missing observations for an additional metric-specific adjustment.

775 Immediate-context and routine analyses

776 Light-source and activity-environment analyses modelled the zero-aware one-hour
777 geometric mean melanopic EDI with fixed-power quasi-Tweedie log-mean models
778 and participant-cluster-robust inference^{71,72}. Additive fixed-site models supplied
779 primary category tests. Category-by-site interaction models supplied site-average
780 category estimates, ratios and site-specific context when the interaction was
781 supported. The activity-environment analysis retained fractional $1/k$ weights for
782 concurrent labels. FDR adjustments were applied to the declared category and site-
783 contrast families. Exact-zero calibration, residual dependence, sparse cells and
784 deletion refits were retained as model checks. A secondary population-mean
785 sensitivity added six participant-level category-proportion terms to the light-source
786 model while retaining participant-clustered covariance. This separated hour-to-hour
787 category differences within participants from between-participant differences in
788 category composition; it did not estimate participant random slopes. Exploratory
789 `glmmTMB` Tweedie log-link models fitted by maximum likelihood then added one
790 common participant intercept to site-by-category fixed effects at the near-eye
791 position for light source and separately at both positions for the activity-environment
792 variable, with the power fixed at 1.539919^{74,75}. The activity-environment models
793 retained the five named categories, excluded Other-only hours and used exact $1/k$
794 weights. Their fixed-predictor variance used the same fractional weights so every
795 participant-hour contributed one unit. Marginal and conditional R^2 used a lognormal
796 distribution-specific variance approximation, and hierarchy-respecting Shapley
797 allocations distributed marginal R^2 across site, category and their interaction. These
798 were model-based point descriptions without bootstrap intervals. No model included
799 random category slopes.

800 The participant-hour analysis modelled each supported participant-hour with a zero-
801 inclusive quasi-Poisson log-link model, fixed site associations and participant-cluster
802 HC3 uncertainty. Free versus work day, active versus sedentary daily activity status
803 and previous-night sleep duration centred at 8 h were entered together. Three-test
804 FDR families were applied separately to pooled associations, predictor-by-site
805 interactions and practical contrasts.

806 **Person-level analyses**

807 Light-related behaviour, visual sensitivity, chronotype, age and measured biological
808 sex were evaluated with site-adjusted metric-specific models. Participant-day
809 outcomes generally used Gaussian or Tweedie mixed models with participant
810 random intercepts; participant-level dynamics outcomes used fixed-effect models.
811 Each analysis retained its complete prespecified false-discovery-rate families,
812 including non-estimable outcomes. Chronotype instruments were modelled
813 separately across five timing outcomes. Age associations were scaled per decade,
814 and female-minus-male contrasts used male as the reference. Average associations
815 and predictor-by-site interactions were distinct questions.

816 The complete biological-sex curve analysis used supported 30-minute arithmetic
817 means transformed as $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$. The generalised additive
818 model included cyclic common and female-specific curves, site curves, participant
819 curves, participant-day random effects and boundary-aware first-order
820 autoregression. The primary test jointly assessed the time-constant and time-varying
821 female-minus-male contributions across the complete 24-hour curve. Clock-specific
822 uncertainty was participant-cluster robust. Level-versus-shape attribution and
823 adjustment for the activity-environment variable were secondary or exploratory
824 analyses.

825 **Multiplicity, uncertainty, model checks and sensitivities**

826 Tests were two-sided. We controlled the false discovery rate with the Benjamini-
827 Hochberg procedure separately within each declared complete family⁷⁸. We report
828 95% confidence intervals and FDR-adjusted p values for confirmatory families. A
829 result that did not meet an FDR-adjusted criterion was treated as inconclusive rather
830 than as evidence of absence or equivalence. Unless stated otherwise, site-average
831 estimates gave each included site equal weight on the fitted scale before back-
832 transformation. Exploratory analyses retained separate multiplicity families and are
833 labelled as such.

834 Model checks addressed convergence, rank, singularity, residual distribution, zero
835 mass, fitted support, heteroscedasticity, serial dependence, nonlinear-term overlap

and participant or site influence. Qualifications were attached to the affected results. Prespecified or structured sensitivities included the separately prepared gap-timing-unaware dataset, common sensor-position samples, alternative support or metric definitions, model-form changes, participant deletion and leave-one-site-out checks. Near-eye and chest models remained separate in every comparison. Analyses were conducted in R, with tidyverse packages supporting data handling and reporting^{61,79}. Exact package versions, formulas, inputs and output files are documented in the Supplementary Information and reproducibility records.

Preregistration and deviations

The study was preregistered with AsPredicted (#273407)⁸⁰. The final analyses preserve the registered scientific questions but include documented changes to placement priority, state and coverage rules, metric definitions, outcome construction, model families, site structure, multiplicity and sensitivity handling. The main manuscript reports the resulting estimands rather than the historical hypothesis sequence. A complete mapping in the Supplementary Information distinguishes scientific deviations and qualifications from analytical history and technical provenance.

Artificial-intelligence assistance

During the original manuscript preparation, generative artificial intelligence tools (ChatGPT 5.5) were used to support language editing, structural refinement and text condensation. AI assistance was used to improve clarity, coherence, concision and journal fit, but **not** to generate, analyse or interpret primary data.

During primary data analysis, AI assistance was used to debug code sections, support figure and table styling, and resolve modelling convergence and singularity issues. Participant comments in their native languages were translated into English with AI assistance. All translations were reviewed and approved or corrected by the respective site authors. AI assistance was **not** used to develop the primary analysis strategy, write analysis scripts or interpret results.

During review, an agentic AI assistant (ChatGPT 5.6 Sol) was used to review the analyses for potential issues. This review prompted additional sensitivity analyses to

evaluate data-preparation and model-selection choices, expanded model diagnostics, and improved effect-size reporting through Shapley decompositions and more precise language. AI assistance also supported expanded analysis documentation and reproducibility checks.

All AI-assisted text was critically reviewed, edited, and approved by the authors, who take full responsibility for the content, accuracy, integrity, and conclusions of the manuscript.

Data availability

All data used in this study are available through the [MeLiDos Project GitHub organisation](#) and are archived in the [MeLiDos Zenodo community](#). Public site datasets have persistent records for Kumasi (GH), Madrid (ES), Dortmund (DE), Izmir (TR), Tübingen (DE), San José (CR), Munich (DE), Borås (SE) and Delft (NL)⁸¹–⁸⁹. These records and other project resources are indexed through the MeLiDos Project Data Hub⁹⁰ and accessed reproducibly through the documented `melidosData` interface⁹¹.

Code availability

All analysis code is available in the [project repository](#), with rendered reports on [GitHub Pages](#) and an archive on [Zenodo](#). The archived version will be reconciled with the final manuscript snapshot before submission.

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1171 Competing interests

1172 M.S. declares the following potential competing interests between 2021 and 2025:
1173 academic roles as a member of the Board of Directors of the Society of Light,
1174 Rhythms, and Circadian Health, Chair of Joint Technical Committee 20 of the
1175 International Commission on Illumination, member of the Daylight Academy, and
1176 Chair of the Research Data Alliance Working Group Optical Radiation and Visual
1177 Experience Data; remunerated roles as Speaker of the Steering Committee of the
1178 Daylight Academy, ad hoc reviewer for the Health and Digital Executive Agency of
1179 the European Commission and the Swedish Research Council, Associate Editor for
1180 *LEUKOS*, examiner for the University of Manchester, Flinders University and the
1181 University of Southern Norway, and consultant for LyS Technologies and RoX
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1190 Stiftung; and being named on European Patent Application EP23159999.4A,
1191 “System and method for corneal-plane physiologically-relevant light logging with an
1192 application to personalized light interventions related to health and well-being.” M.S.
1193 declares that these roles and relationships had no influence on the work presented
1194 here.

1195 J.Z. declares the following potential competing interests between 2021 and 2025:
1196 academic roles as a member of Joint Technical Committee 20 of the International
1197 Commission on Illumination, member of the Research Data Alliance Working Group
1198 Optical Radiation and Visual Experience Data, and Speaker of group 2, melanopic
1199 effects of light, of the Technical Scientific Committee of the German Society of
1200 Lighting Technology and Design; remunerated roles as examiner for the Swiss
1201 Lighting Society; teacher for the German Society of Lighting Technology and Design,
1202 Munich University of Applied Sciences and the Technical University of Applied
1203 Sciences Rosenheim; associated partner at 3lpi lighting design + engineering,
1204 Munich; tool and three-dimensional-model designer for Zumtobel Lighting GmbH;
1205 and course designer for Munich University of Applied Sciences and Virtual University
1206 Bavaria; honoraria for talks from the German Society of Lighting Technology and
1207 Design, Lamilux/Heinrich Strunz GmbH, Robert Bosch Hospital Stuttgart, Ergotopia
1208 GmbH, the German Social Accident Insurance Institution for the Administrative
1209 Sector, BRIKEN CULTUR, KITEO GmbH & Co. KG and Augsburg University of
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1211 together with 3lpi, holding European Union Intellectual Property Office design patent
1212 008194021-0001 through -0006 for a non-visually optimized luminaire.

1213 A.D. declares academic roles as a member of Joint Technical Committee 20 of the
1214 International Commission on Illumination and Division Reporter DR6-50 for the “4th
1215 Manchester Workshop on Light Metrics for Biology: Light Pollution.” A.D. declares
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1217 Minus 15 B.V., a company active in the lighting sector. All other authors declare no
1218 competing interests.

219	Supplementary Information
220	Study support and exposure definitions
221	Supplementary Table S1. Descriptive sample flow

Descriptive sample flow and analytical support		
Stage	Quantity	Count
Participant roster		
Available normalized participant metadata	Participants	191
Primary near-eye sensor position		
At least 80% complete before all-zero screen	Participant-days	818
Exact all-zero days excluded	Participant-days	2
Final descriptive dataset	Participants	141
Final descriptive dataset	Participant-days	816
Final descriptive dataset	One-minute real observations	1,175,160
Complementary chest sensor position		
At least 80% complete before all-zero screen	Participant-days	905
Exact all-zero days excluded	Participant-days	3
Final descriptive dataset	Participants	154
Final descriptive dataset	Participant-days	902
Final descriptive dataset	One-minute real observations	1,298,880
Paired main subset		
Paired main subset	Participants	112
Paired main subset	Paired participant-days	643
Counts use their stated denominator. An excluded participant-day is an all-zero day, not necessarily an excluded participant. The paired row reports paired participant-days, not minute-level observation pairs.		

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223 **Supplementary Table S1.** Participant, participant-day, and one-minute observation counts across the roster, sensor-specific completeness and
224 all-zero screens, final descriptive datasets, and paired main subset. Each row names its denominator explicitly. An excluded participant-day is not
225 necessarily an excluded participant.

226 **Supplementary Table S2. Full near-eye metric dictionary and distributions**

Metric descriptive summary (near eye)							
Metric [†]	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)
Duration							
Time above 1,000 lx melEDI Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	HH:MM	00:41 (00:11, 01:33) <i>01:10 ± 01:28</i> N=141; d=816	01:18 (00:31, 03:24) <i>02:00 ± 01:54</i> N=13; d=78	01:18 (00:29, 02:12) <i>01:33 ± 01:28</i> N=13; d=78	01:03 (00:20, 02:32) <i>01:48 ± 02:14</i> N=18; d=107	00:40 (00:07, 01:25) <i>01:06 ± 01:22</i> N=26; d=150	00:39 (00:10, 01:46) <i>01:09 ± 01:18</i> N=10; d=60
Time above 250 lx melEDI during wake Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	HH:MM	02:25 (00:51, 04:37) <i>03:00 ± 02:29</i> N=141; d=737	04:06 (02:30, 06:15) <i>04:21 ± 02:24</i> N=13; d=73	03:29 (02:03, 05:16) <i>03:39 ± 02:25</i> N=13; d=64	03:42 (02:05, 05:58) <i>04:05 ± 02:51</i> N=18; d=90	02:01 (00:47, 04:20) <i>02:52 ± 02:39</i> N=26; d=141	02:38 (00:55, 04:19) <i>02:57 ± 02:19</i> N=10; d=54
Time below 10 lx melEDI before sleep Low-light time before bed; limits evening melatonin suppression and circadian delay.	HH:MM	01:53 (01:01, 02:35) <i>01:51 ± 01:02</i> N=139; d=655	02:16 (01:30, 02:39) <i>02:05 ± 00:53</i> N=13; d=67	01:50 (01:21, 02:52) <i>01:59 ± 00:57</i> N=12; d=50	01:42 (00:53, 02:17) <i>01:41 ± 00:58</i> N=18; d=84	01:52 (01:04, 02:25) <i>01:52 ± 01:06</i> N=26; d=132	01:43 (00:42, 02:32) <i>01:45 ± 01:10</i> N=10; d=52
Time below 1 lx melEDI during sleep Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	HH:MM	07:08 (05:58, 08:15) <i>07:04 ± 01:57</i> N=141; d=778	07:08 (06:03, 08:18) <i>07:12 ± 01:46</i> N=13; d=76	07:14 (05:56, 08:06) <i>07:01 ± 01:50</i> N=13; d=68	07:08 (05:56, 08:16) <i>07:05 ± 01:50</i> N=18; d=95	06:54 (05:51, 07:46) <i>06:47 ± 01:30</i> N=26; d=150	05:52 (04:41, 07:03) <i>05:58 ± 01:58</i> N=10; d=60
Longest period above 250 lx melEDI Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	HH:MM	00:38 (00:17, 01:12) <i>00:55 ± 01:00</i> N=141; d=816	00:57 (00:28, 01:27) <i>01:12 ± 01:02</i> N=13; d=78	00:48 (00:23, 01:20) <i>01:09 ± 01:15</i> N=13; d=78	01:00 (00:21, 01:40) <i>01:20 ± 01:31</i> N=18; d=107	00:35 (00:13, 01:06) <i>00:50 ± 00:54</i> N=26; d=150	00:38 (00:17, 01:32) <i>01:01 ± 01:02</i> N=10; d=60
Dynamics							
Interdaily stability Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	—	0.308 (0.248, 0.38) <i>0.318 ± 0.095</i> N=141; d=816	0.311 (0.272, 0.367) <i>0.315 ± 0.078</i> N=13; d=78	0.258 (0.239, 0.389) <i>0.288 ± 0.094</i> N=13; d=78	0.264 (0.219, 0.319) <i>0.274 ± 0.074</i> N=18; d=107	0.329 (0.263, 0.424) <i>0.344 ± 0.107</i> N=26; d=150	0.244 (0.218, 0.277) <i>0.252 ± 0.058</i> N=10; d=60
Intradaily variability Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	—	1.253 (0.93, 1.502) <i>1.229 ± 0.389</i> N=141; d=816	0.997 (0.818, 1.388) <i>1.088 ± 0.343</i> N=13; d=78	1.193 (0.96, 1.389) <i>1.238 ± 0.345</i> N=13; d=78	1.182 (0.928, 1.49) <i>1.177 ± 0.407</i> N=18; d=107	1.154 (0.801, 1.442) <i>1.125 ± 0.354</i> N=26; d=150	1.48 (0.869, 1.642) <i>1.332 ± 0.477</i> N=10; d=60
Exposure history							
melEDI dose Intensity–duration-weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	klx·h	4.96 (1.936, 12.313) <i>10.358 ± 16.216</i> N=141; d=761	9.296 (3.892, 21.895) <i>19.923 ± 27.784</i> N=13; d=71	10.495 (3.855, 17.395) <i>15.881 ± 20.395</i> N=13; d=74	6.751 (2.469, 14.435) <i>13.809 ± 23.711</i> N=18; d=97	3.862 (1.411, 10.217) <i>7.719 ± 9.847</i> N=26; d=137	7.11 (1.897, 15.789) <i>10.055 ± 11.237</i> N=10; d=55

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Table continued, column panel A

Metric descriptive summary (near eye)							
Metric [†]	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)
Level							
Mean melEDI Geometric average of daily melEDI values, including zeros; summarizes overall exposure while reducing peak influence.	lx	5.154 (2.831, 9.225) <i>7.587 ± 9.496</i> N=141; d=816	7.569 (4.55, 11.613) <i>9.171 ± 7.111</i> N=13; d=78	6.659 (4.18, 11.674) <i>8.274 ± 6.558</i> N=13; d=78	7.704 (3.081, 13.924) <i>12.017 ± 18.865</i> N=18; d=107	5.232 (3.224, 8.188) <i>7.221 ± 6.33</i> N=26; d=150	9.013 (4.011, 15.089) <i>12.102 ± 12.013</i> N=10; d=60
Brightest 10 h mean Mean of the brightest 10 hours; reflects the strength of the main daytime light episode.	lx	110.566 (41.513, 243.212) <i>214.866 ± 508.56</i> N=141; d=816	252.689 (104.514, 461.953) <i>399.302 ± 518.102</i> N=13; d=78	175.974 (78.625, 299.106) <i>262.206 ± 329.201</i> N=13; d=78	156.184 (67.783, 339.731) <i>422.54 ± 1,209.616</i> N=18; d=107	100.057 (33.255, 209.904) <i>179.372 ± 241.53</i> N=26; d=150	130.78 (54.044, 259.726) <i>201.947 ± 194.042</i> N=10; d=60
Darkest 10 h mean Mean of the darkest 10 hours; lower values during the biological night favour melatonin preservation and sleep.	lx	0.103 (0.02, 0.253) <i>0.243 ± 0.513</i> N=141; d=816	0.08 (0.016, 0.191) <i>0.183 ± 0.299</i> N=13; d=78	0.1 (0.035, 0.23) <i>0.2 ± 0.332</i> N=13; d=78	0.14 (0.036, 0.282) <i>0.223 ± 0.299</i> N=18; d=107	0.149 (0.074, 0.309) <i>0.273 ± 0.465</i> N=26; d=150	0.325 (0.119, 0.71) <i>0.515 ± 0.653</i> N=10; d=60
Spectrum							
Melanopic daylight efficacy ratio Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	—	0.724 (0.643, 0.795) <i>0.724 ± 0.116</i> N=137; d=687	0.788 (0.725, 0.872) <i>0.796 ± 0.162</i> N=13; d=74	0.743 (0.699, 0.796) <i>0.743 ± 0.077</i> N=13; d=65	0.781 (0.675, 0.853) <i>0.768 ± 0.118</i> N=18; d=97	0.65 (0.592, 0.728) <i>0.667 ± 0.098</i> N=26; d=144	0.724 (0.652, 0.773) <i>0.72 ± 0.093</i> N=10; d=55

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Table continued, column panel A

Metric descriptive summary (near eye)							
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)
Timing							
Midpoint of the brightest 10 hours Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	HH:MM clock time	13:44 (12:48, 15:00) <i>13:56 ± 01:51</i> N=141; d=816	13:16 (12:28, 14:10) <i>13:20 ± 01:33</i> N=13; d=78	14:12 (13:23, 15:18) <i>14:22 ± 01:51</i> N=13; d=78	13:52 (12:54, 15:04) <i>14:00 ± 01:42</i> N=18; d=107	13:38 (12:47, 14:44) <i>13:52 ± 01:42</i> N=26; d=150	14:54 (12:50, 16:03) <i>14:31 ± 02:14</i> N=10; d=60
Midpoint of the darkest 10 hours Centre time of the darkest 10 hours; indexes the main daily darkness cue.	HH:MM clock time	02:54 (02:01, 03:50) <i>02:58 ± 01:41</i> N=141; d=816	02:02 (01:25, 02:48) <i>02:03 ± 01:07</i> N=13; d=78	02:55 (02:01, 03:27) <i>02:50 ± 02:05</i> N=13; d=78	02:32 (01:45, 03:22) <i>02:37 ± 01:19</i> N=18; d=107	03:08 (02:11, 04:10) <i>03:16 ± 01:34</i> N=26; d=150	02:37 (01:54, 04:17) <i>03:10 ± 02:00</i> N=10; d=60
First light timing above 250 lx melEDI First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	HH:MM clock time	09:08 (08:02, 10:40) <i>09:26 ± 02:22</i> N=140; d=727	07:57 (06:56, 08:33) <i>08:03 ± 01:32</i> N=13; d=74	09:04 (08:18, 09:52) <i>09:17 ± 01:56</i> N=13; d=68	08:36 (07:10, 10:13) <i>08:45 ± 02:21</i> N=18; d=93	09:43 (08:13, 10:53) <i>09:45 ± 02:21</i> N=25; d=133	09:36 (07:58, 10:53) <i>09:39 ± 03:12</i> N=10; d=56
Last light timing above 250 lx melEDI Last bright-light exposure; later timing may delay circadian phase and sleep onset.	HH:MM clock time	18:08 (16:27, 19:42) <i>18:03 ± 02:36</i> N=141; d=687	18:41 (17:38, 19:38) <i>18:39 ± 01:38</i> N=13; d=73	18:30 (17:37, 19:59) <i>18:37 ± 02:08</i> N=13; d=62	19:34 (18:04, 20:42) <i>19:16 ± 02:28</i> N=18; d=89	17:37 (15:35, 19:09) <i>17:18 ± 02:44</i> N=26; d=121	19:50 (18:37, 20:50) <i>19:33 ± 02:11</i> N=10; d=53
Mean timing of exposure above 250 lx melEDI Average bright-light timing; summarizes the phase of daily circadian stimulation.	HH:MM clock time	13:29 (12:29, 14:39) <i>13:31 ± 01:52</i> N=141; d=742	13:20 (12:29, 14:24) <i>13:21 ± 01:25</i> N=13; d=72	13:52 (13:13, 15:06) <i>14:02 ± 01:41</i> N=13; d=71	13:44 (12:28, 14:55) <i>13:51 ± 01:56</i> N=18; d=93	13:17 (12:17, 14:31) <i>13:24 ± 01:45</i> N=26; d=138	14:04 (12:58, 15:25) <i>14:09 ± 02:22</i> N=10; d=55
Median (25th percentile, 75th percentile), <i>circular or arithmetic mean ± standard deviation</i> , N=participants; d=participant-days with a finite value for that metric.							
¹ The brief meaning and relevance notes describe established physiological constructs; this descriptive table does not estimate individual health effects.							
² Scaling describes the distribution column only; printed values remain on their stated scale. Symlog uses base 10 with a linear region through 1 in the stated unit; Identity is linear; Circular clock unwraps values around the metric-specific clock centre.							
³ Red lines indicate site medians. Symlog and circular-clock rows are transformed only for plotting.							

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Table continued, column panel A

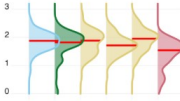
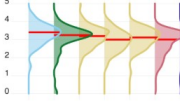
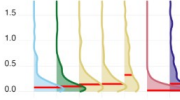
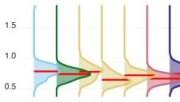
Metric descriptive summary (near eye)								
Metric ¹	Unit	Overall	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Duration								
Time above 1,000 lx melEDI Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	HH:MM	00:41 (00:11, 01:33) 01:10 ± 01:28 N=141; d=816	00:27 (00:06, 00:53) 00:38 ± 00:47 N=23; d=129	00:30 (00:06, 01:04) 00:44 ± 00:51 N=17; d=101	00:19 (00:09, 00:43) 00:32 ± 00:33 N=6; d=32	00:43 (00:17, 01:14) 00:58 ± 00:57 N=15; d=81	Symlog (base 10; threshold 1)	
Time above 250 lx melEDI during wake Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	HH:MM	02:25 (00:51, 04:37) 03:00 ± 02:29 N=141; d=737	02:43 (00:54, 04:43) 03:02 ± 02:19 N=23; d=113	01:30 (00:41, 03:26) 02:09 ± 01:57 N=17; d=94	01:39 (00:33, 02:40) 01:48 ± 01:21 N=6; d=30	01:14 (00:25, 02:37) 01:45 ± 01:46 N=15; d=78	Symlog (base 10; threshold 1)	
Time below 10 lx melEDI before sleep Low-light time before bed; limits evening melatonin suppression and circadian delay.	HH:MM	01:53 (01:01, 02:35) 01:51 ± 01:02 N=139; d=655	01:43 (00:58, 02:50) 01:50 ± 01:05 N=23; d=105	01:31 (00:49, 02:15) 01:38 ± 01:00 N=17; d=82	02:16 (01:15, 02:39) 01:56 ± 01:00 N=6; d=25	02:30 (01:28, 02:52) 02:09 ± 00:57 N=14; d=58	Symlog (base 10; threshold 1)	
Time below 1 lx melEDI during sleep Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	HH:MM	07:08 (05:58, 08:15) 07:04 ± 01:57 N=141; d=778	08:06 (07:10, 09:10) 08:02 ± 01:57 N=23; d=118	06:45 (05:08, 08:15) 06:40 ± 02:07 N=17; d=99	06:56 (06:06, 07:36) 06:39 ± 01:43 N=6; d=31	07:31 (06:31, 09:00) 07:35 ± 02:15 N=15; d=81	Symlog (base 10; threshold 1)	
Longest period above 250 lx melEDI Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	HH:MM	00:38 (00:17, 01:12) 00:55 ± 01:00 N=141; d=816	00:38 (00:16, 01:04) 00:45 ± 00:36 N=23; d=129	00:34 (00:14, 00:55) 00:43 ± 00:41 N=17; d=101	00:28 (00:16, 00:43) 00:30 ± 00:20 N=6; d=32	00:23 (00:15, 00:49) 00:38 ± 00:45 N=15; d=81	Symlog (base 10; threshold 1)	
Dynamics								
Interdaily stability Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	—	0.308 (0.248, 0.38) 0.318 ± 0.095 N=141; d=816	0.352 (0.321, 0.455) 0.368 ± 0.092 N=23; d=129	0.278 (0.254, 0.394) 0.323 ± 0.104 N=17; d=101	0.342 (0.315, 0.473) 0.37 ± 0.118 N=6; d=32	0.293 (0.255, 0.354) 0.299 ± 0.062 N=15; d=81	Identity	
Intradaily variability Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	—	1.253 (0.93, 1.502) 1.229 ± 0.389 N=141; d=816	1.251 (0.999, 1.497) 1.251 ± 0.413 N=23; d=129	1.359 (0.95, 1.546) 1.307 ± 0.415 N=17; d=101	1.399 (1.282, 1.483) 1.355 ± 0.256 N=6; d=32	1.324 (1.029, 1.595) 1.344 ± 0.413 N=15; d=81	Identity	
Exposure history								
melEDI dose Intensity–duration–weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	klx·h	4.96 (1.936, 12.313) 10.358 ± 16.216 N=141; d=761	3.317 (1.703, 6.29) 6.115 ± 8.79 N=23; d=125	4.35 (1.791, 10.625) 7.49 ± 8.898 N=17; d=96	2.195 (1.339, 4.117) 3.281 ± 2.573 N=6; d=30	6.045 (2.49, 14.907) 10.016 ± 10.799 N=15; d=76	Symlog (base 10; threshold 1)	

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Table continued, column panel A

Metric descriptive summary (near eye)								
Metric ¹	Unit	Overall	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Level								
Mean melEDI Geometric average of daily melEDI values, including zeros; summarizes overall exposure while reducing peak influence.	lx	5.154 (2.831, 9.225) <i>7.587 ± 9.496</i> N=141; d=816	3.505 (1.469, 6.211) <i>4.166 ± 3.272</i> N=23; d=129	5.857 (3.726, 7.986) <i>7.778 ± 6.946</i> N=17; d=101	4.325 (3.585, 6.77) <i>5.248 ± 2.85</i> N=6; d=32	2.144 (0.744, 4.042) <i>3.019 ± 2.896</i> N=15; d=81	Symlog (base 10; threshold 1)	
Brightest 10 h mean Mean of the brightest 10 hours; reflects the strength of the main daytime light episode.	lx	110.566 (41.513, 243.212) <i>214.866 ± 508.56</i> N=141; d=816	101.102 (24.355, 218.457) <i>136.312 ± 134.07</i> N=23; d=129	83.24 (53.17, 151.786) <i>120.172 ± 101.545</i> N=17; d=101	68.143 (45.331, 132.458) <i>96.735 ± 72.311</i> N=6; d=32	46.66 (9.242, 103.175) <i>82.489 ± 115.07</i> N=15; d=81	Symlog (base 10; threshold 1)	
Darkest 10 h mean Mean of the darkest 10 hours; lower values during the biological night favour melatonin preservation and sleep.	lx	0.103 (0.02, 0.253) <i>0.243 ± 0.513</i> N=141; d=816	0.022 (0, 0.08) <i>0.054 ± 0.075</i> N=23; d=129	0.152 (0.052, 0.481) <i>0.475 ± 0.882</i> N=17; d=101	0.133 (0.055, 0.309) <i>0.411 ± 1.044</i> N=6; d=32	0.01 (0, 0.092) <i>0.059 ± 0.104</i> N=15; d=81	Symlog (base 10; threshold 1)	
Spectrum								
Melanopic daylight efficacy ratio Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	—	0.724 (0.643, 0.795) <i>0.724 ± 0.116</i> N=137; d=687	0.668 (0.629, 0.721) <i>0.673 ± 0.07</i> N=22; d=83	0.747 (0.634, 0.834) <i>0.735 ± 0.122</i> N=17; d=91	0.719 (0.661, 0.761) <i>0.706 ± 0.067</i> N=6; d=30	0.755 (0.729, 0.82) <i>0.758 ± 0.097</i> N=12; d=48	Identity	

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Table continued, column panel A

Metric descriptive summary (near eye)								
Metric ¹	Unit	Overall	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Timing								
Midpoint of the brightest 10 hours Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	HH:MM clock time	13:44 (12:48, 15:00) 13:56 ± 01:51 N=141; d=816	13:44 (13:06, 15:20) 14:10 ± 01:51 N=23; d=129	14:10 (13:32, 15:21) 14:25 ± 01:37 N=17; d=101	12:34 (11:49, 14:02) 12:55 ± 01:44 N=6; d=32	12:49 (12:06, 13:39) 13:03 ± 01:58 N=15; d=81	Circular clock	
Midpoint of the darkest 10 hours Centre time of the darkest 10 hours; indexes the main daily darkness cue.	HH:MM clock time	02:54 (02:01, 03:50) 02:58 ± 01:41 N=141; d=816	03:39 (02:43, 04:34) 03:34 ± 01:49 N=23; d=129	03:31 (02:33, 04:14) 03:26 ± 01:25 N=17; d=101	01:45 (01:05, 03:01) 02:10 ± 01:27 N=6; d=32	02:33 (02:04, 03:17) 02:36 ± 01:36 N=15; d=81	Circular clock	
First light timing above 250 lx melEDI First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	HH:MM clock time	09:08 (08:02, 10:40) 09:26 ± 02:22 N=140; d=727	09:32 (08:40, 11:21) 10:05 ± 02:18 N=23; d=118	10:01 (09:06, 11:52) 10:26 ± 02:28 N=17; d=95	08:11 (07:26, 09:57) 08:36 ± 01:30 N=6; d=29	08:43 (08:01, 10:10) 09:19 ± 01:52 N=15; d=61	Circular clock	
Last light timing above 250 lx melEDI Last bright-light exposure; later timing may delay circadian phase and sleep onset.	HH:MM clock time	18:08 (16:27, 19:42) 18:03 ± 02:36 N=141; d=687	17:48 (16:46, 19:19) 18:01 ± 02:21 N=23; d=106	18:45 (17:02, 19:38) 18:23 ± 02:15 N=17; d=89	15:45 (13:20, 16:38) 15:22 ± 02:38 N=6; d=28	15:54 (14:34, 17:25) 15:46 ± 02:13 N=15; d=66	Circular clock	
Mean timing of exposure above 250 lx melEDI Average bright-light timing; summarizes the phase of daily circadian stimulation.	HH:MM clock time	13:29 (12:29, 14:39) 13:31 ± 01:52 N=141; d=742	13:43 (12:55, 14:29) 13:47 ± 01:33 N=23; d=117	13:58 (12:54, 14:58) 14:04 ± 01:39 N=17; d=94	11:30 (10:28, 12:38) 11:26 ± 01:36 N=6; d=31	12:17 (11:04, 13:04) 12:11 ± 01:36 N=15; d=71	Circular clock	
Median (25th percentile, 75th percentile), <i>circular or arithmetic mean ± standard deviation</i> , N=participants; d=participant-days with a finite value for that metric.								
¹ The brief meaning and relevance notes describe established physiological constructs; this descriptive table does not estimate individual health effects.								
² Scaling describes the distribution column only; printed values remain on their stated scale. Symlog uses base 10 with a linear region through 1 in the stated unit; Identity is linear; Circular clock unwraps values around the metric-specific clock centre.								
³ Red lines indicate site medians. Symlog and circular-clock rows are transformed only for plotting.								

Supplementary Table S2. Definitions, exact support, median (interquartile range) summaries, and density displays for the 17 near-eye personal light-exposure metrics overall and by site. Values use three decimal places where applicable. Metrics are distinct summaries and should not be interpreted as interchangeable outcomes.

Site ¹	Minutes in the recommended range: ²				Share of all eligible real minutes: ³			
	Daytime	Pre-sleep	Sleep ⁴	Total ⁵	Daytime	Pre-sleep	Sleep	Unclassified ⁶
Overall	24.0% 137,792 / 573,712	63.3% 81,894 / 129,390	87.7% 336,052 / 383,366	51.2% 555,738 / 1,086,466	51.6% 606,215 / 1,175,160	11.9% 140,152 / 1,175,160	32.6% 383,607 / 1,175,160	3.8% 45,186 / 1,175,160
Borås (SE)	33.2% 19,225 / 57,952	69.9% 8,914 / 12,759	93.4% 33,152 / 35,489	57.7% 61,291 / 106,200	54.0% 60,642 / 112,260	12.2% 13,704 / 112,260	31.7% 35,549 / 112,260	2.1% 2,365 / 112,260
Delft (NL)	29.6% 14,760 / 49,852	66.7% 7,181 / 10,774	86.4% 29,915 / 34,621	54.4% 51,856 / 95,247	47.0% 52,755 / 112,260	10.9% 12,283 / 112,260	30.9% 34,682 / 112,260	11.2% 12,540 / 112,260
Dortmund (DE)	32.4% 22,631 / 69,900	59.6% 9,749 / 16,361	87.5% 42,829 / 48,928	55.6% 75,209 / 135,189	48.1% 74,094 / 154,080	11.3% 17,409 / 154,080	31.8% 48,928 / 154,080	8.9% 13,649 / 154,080
Tübingen (DE)	22.5% 25,082 / 111,396	61.8% 15,623 / 25,283	87.1% 61,030 / 70,029	49.2% 101,735 / 206,708	55.1% 119,150 / 216,120	12.4% 26,881 / 216,120	32.4% 70,089 / 216,120	0.0% 0 / 216,120
Munich (DE)	25.0% 10,681 / 42,685	60.6% 5,963 / 9,833	71.7% 21,485 / 29,951	46.2% 38,129 / 82,469	53.0% 45,831 / 86,400	12.3% 10,618 / 86,400	34.7% 29,951 / 86,400	0.0% 0 / 86,400
Madrid (ES)	24.2% 21,067 / 87,030	62.7% 12,766 / 20,366	95.9% 58,374 / 60,891	54.8% 92,207 / 168,287	48.4% 89,948 / 185,880	11.7% 21,684 / 185,880	32.8% 60,891 / 185,880	7.2% 13,357 / 185,880
Izmir (TR)	17.8% 12,708 / 71,562	55.1% 9,221 / 16,736	79.0% 39,936 / 50,556	44.6% 61,865 / 138,854	51.5% 74,926 / 145,440	12.3% 17,883 / 145,440	34.8% 50,556 / 145,440	1.4% 2,075 / 145,440
San José (CR)	13.8% 3,266 / 23,640	64.6% 3,124 / 4,839	87.1% 12,438 / 14,275	44.0% 18,828 / 42,754	54.9% 25,285 / 46,080	11.5% 5,320 / 46,080	31.0% 14,275 / 46,080	2.6% 1,200 / 46,080
Kumasi (GH)	14.0% 8,372 / 59,695	75.2% 9,353 / 12,439	95.5% 36,893 / 38,626	49.3% 54,618 / 110,760	54.5% 63,584 / 116,640	12.3% 14,370 / 116,640	33.2% 38,686 / 116,640	0.0% 0 / 116,640

¹ Near eye participant and participant-day sample sizes are reported in the participant table; every percentage cell gives its exact minute denominator.

² Recommended ranges follow Brown et al. (2022): Daytime ≥ 250 lx melanopic EDI, Pre-sleep ≤ 10 lx melanopic EDI, and Sleep ≤ 1 lx melanopic EDI. Daytime, Pre-sleep, and Sleep identify the recommendation windows. Each grey numerator/denominator gives minutes within the applicable recommendation range over all valid one-minute observations in that window. Sleep describes the bedside sleep environment.

³ Each grey relevant / all count gives minutes in the displayed diary state over all eligible real minutes before classification by diary state or measurement availability.

⁴ During diary-defined sleep the bedside sensor describes the sleep environment rather than direct ocular exposure.

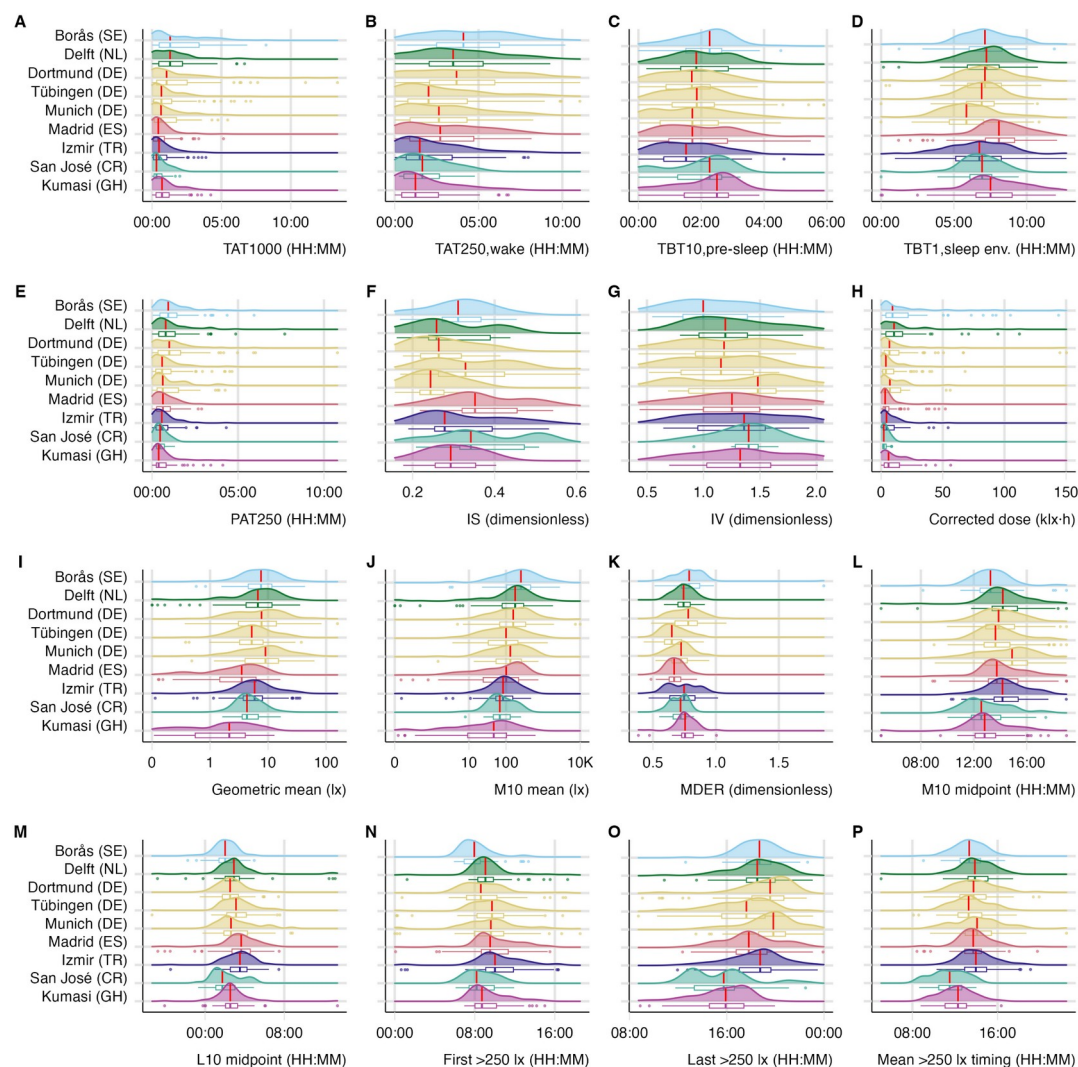
⁵ Total pools all classified valid minutes.

⁶ Unclassified time can result from missing diary state, missing or removed melEDI measurements, or both.

238 **Supplementary Table S3.** Fractions of valid near-eye minutes meeting each Brown et al. recommendation range, with explicit numerators and
239 denominators. Daytime, Pre-sleep, and Sleep identify the recommendation windows. The Sleep summaries use near-eye illuminance recorded
240 during sleep and describe the bedside sleep environment rather than verified ocular exposure. These pooled-minute summaries are descriptive
241 and differ from the modelled recommendation-window-period estimates in main Table 2.

1242 **Supplementary Figure S1. Near-eye metric distributions**

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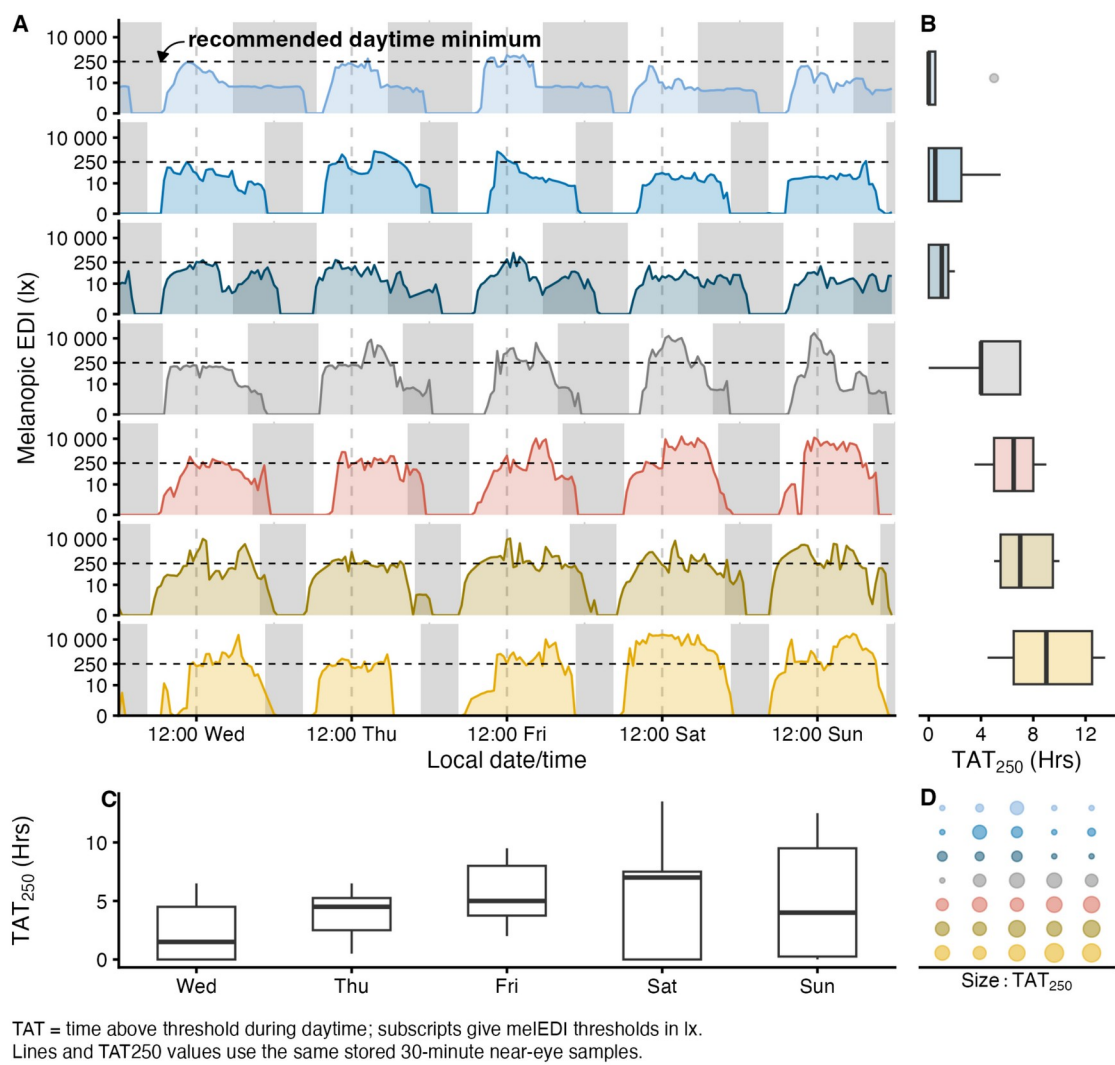


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1245 **Supplementary Figure S1. Detailed near-eye metric distributions by country-coded**
1246 **study site.**

1247 **Supplementary Figure S2. Worked metric derivation**

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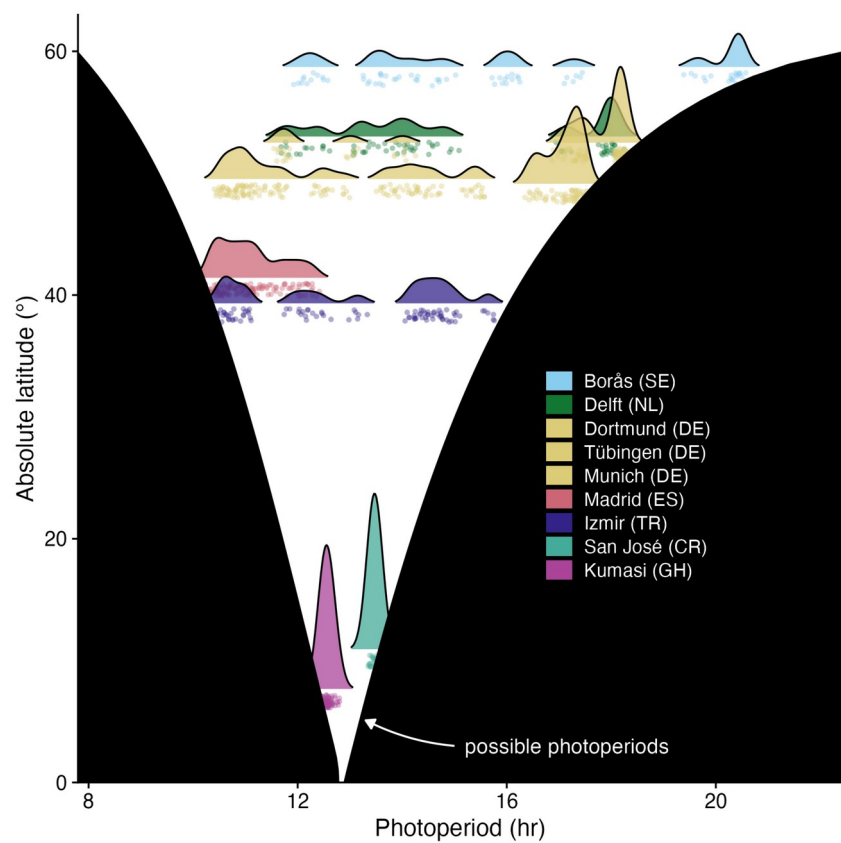


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1250 Supplementary Figure S2. Examples linking 30-minute time series to the daily time-
1251 above-250-lx metric.

1252 **Supplementary Figure S3. Latitude and civil-photoperiod**
1253 **measurement range**

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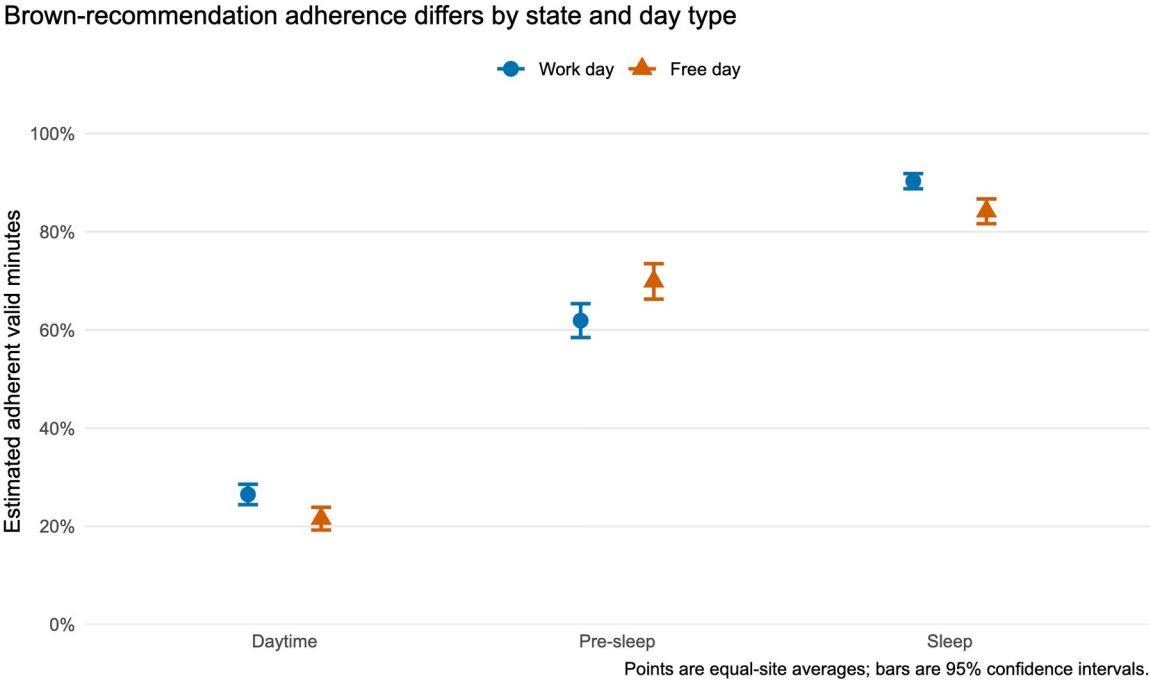


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1256 Supplementary Figure S3. Observed civil photoperiod and theoretical bounds by
1257 absolute latitude.

Recommendation adherence

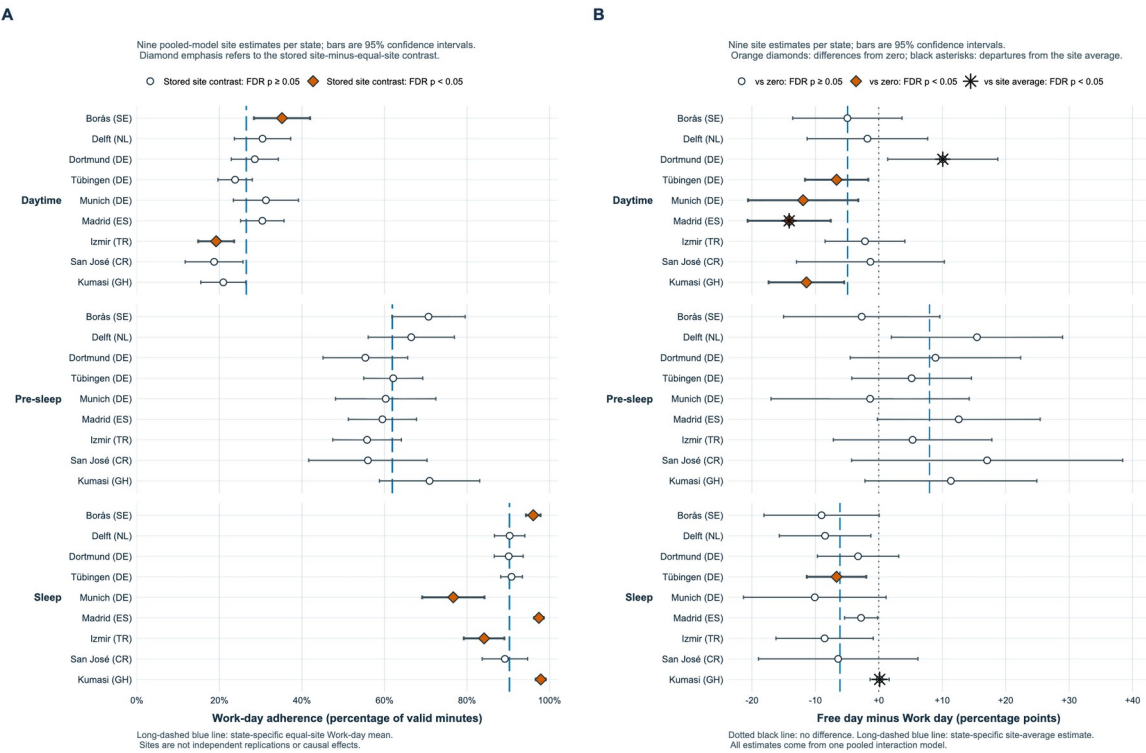
Daytime, Pre-sleep, and Sleep identify the Brown et al. recommendation windows.

Supplementary Figure S4. Site-average recommendation adherence



Supplementary Figure S4. Recommendation adherence in the Daytime, Pre-sleep and Sleep recommendation windows on Work and Free days. Points are equal-site average percentages of valid minutes meeting the Brown et al. recommendation; bars are 95% confidence intervals. Daytime, Pre-sleep and Sleep identify the three Brown et al. recommendation windows. The Sleep estimate uses near-eye illuminance recorded during sleep and describes the bedside sleep environment rather than verified ocular exposure.

Supplementary Figure S5. Site-specific adherence and day-type differences



Supplementary Figure S5. Site-specific recommendation adherence across Brown et al. recommendation windows. A, Work-day adherence, with diamonds identifying deviations from the window-specific site-average Work-day estimate that met its false-discovery-rate (FDR) criterion. B, Free-minus-Work differences, with diamonds identifying differences from zero and asterisks identifying departures from the window-specific site-average difference in a separate 27-comparison FDR family. Daytime, Pre-sleep and Sleep identify the Brown et al. recommendation windows. Points are estimates and bars are 95% confidence intervals. Sites are components of one pooled model, not independent replications or causal effects of location. The Sleep-window estimates use near-eye illuminance recorded during sleep and describe the bedside sleep environment rather than verified ocular exposure.

285 **Supplementary Table S4. Exploratory cross-window recommendation-adherence associations**

Exploratory cross-window recommendation-adherence associations				
Outcome window	Difference per +10 percentage points Daytime adherence, percentage points (95% CI)	FDR-adjusted p	FDR-correction set/family	Interpretation for recommendation adherence
Within participant, linked cycle				
Sleep	-0.26 (-0.95 to 0.43)	0.453	Four prespecified primary cross-window tests	No retained evidence that a cycle with 10 percentage points higher Daytime adherence had different Sleep adherence; the day-level claim is withheld because serial dependence remains unresolved.
Pre-sleep	-1.09 (-2.75 to 0.56)	0.258	Four prespecified primary cross-window tests	No retained evidence that a cycle with 10 percentage points higher Daytime adherence had different Pre-sleep adherence; the day-level claim is withheld because serial dependence remains unresolved.

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Table continued

Exploratory cross-window recommendation-adherence associations				
Outcome window	Difference per +10 percentage points Daytime adherence, percentage points (95% CI)	FDR-adjusted p	FDR-correction set/family	Interpretation for recommendation adherence
Between participants, observed monitoring-period average				
Sleep	-2.52 (-3.55 to -1.48)	<0.001	Four prespecified primary cross-window tests	Participants with 10 percentage points higher average Daytime adherence had 2.52 percentage points lower average Sleep adherence over the monitoring period.
Pre-sleep	-3.59 (-5.94 to -1.25)	0.005	Four prespecified primary cross-window tests	Participants with 10 percentage points higher average Daytime adherence had 3.59 percentage points lower average Pre-sleep adherence over the monitoring period.
All four tests form one FDR-correction set. Site-standardized estimates give nine sites equal weight and use 50:50 Work/Free weighting. The primary sample contained 1,376 target-window rows from 140 participants and 761 linked cycles. Within-participant claims are withheld because temporal dependence remains unresolved. Between-participant estimates describe observed monitoring-period averages, not rankings, stable traits, or causal effects. Confidence intervals are not multiplicity adjusted.				

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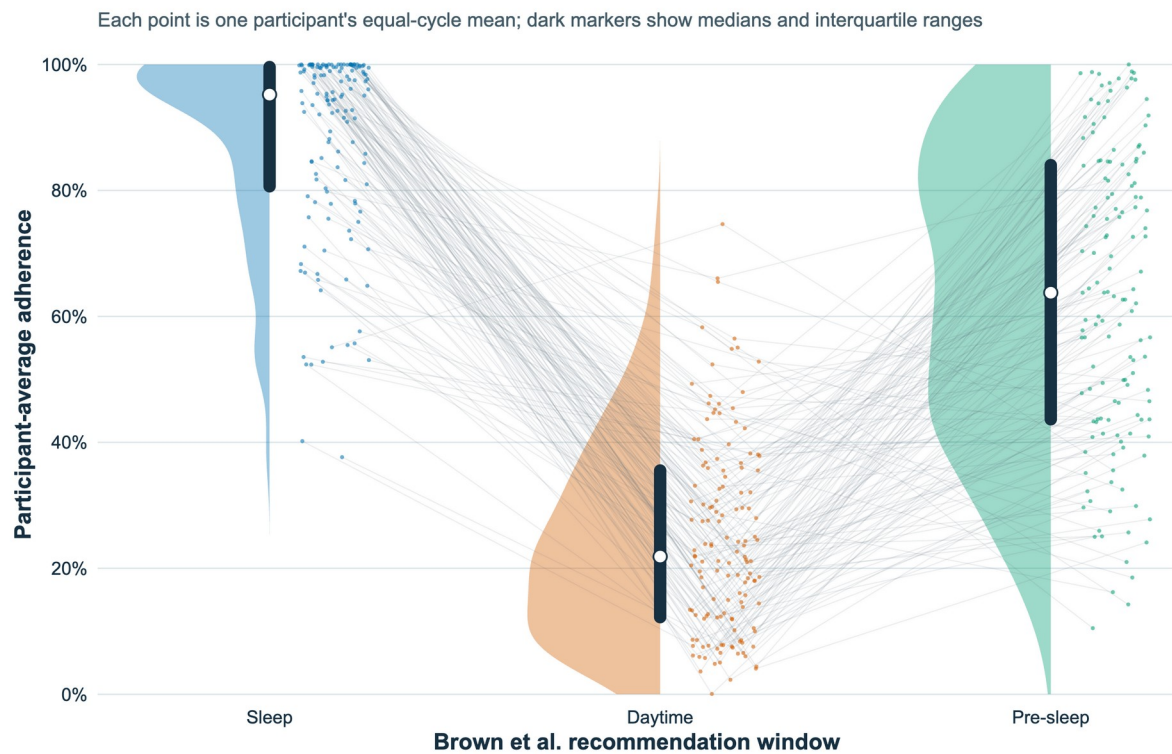
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Supplementary Table S4. Exploratory within-participant and between-participant associations across Brown et al. recommendation windows. Results are percentage-point differences per 10 percentage points higher Daytime adherence. The four tests form one FDR family. The within-

290 participant day-level claim is withheld because serial dependence remains unresolved. Between-participant results describe observed monitoring-
291 period averages and do not establish rankings, stable traits, or causal effects.

Supplementary Figure S6. Anonymous participant recommendation-adherence profiles



Supplementary Figure S6. Participant-average recommendation adherence across the Brown et al. recommendation windows. Daytime, Pre-sleep, and Sleep are the window labels. Each colored point is one anonymous participant's equal-cycle mean within that window. Faint lines connect the same anonymous participant across the three window summaries; they are not time trajectories or ranks. Half-violin shapes show the window distributions, and dark bars with white points show interquartile ranges and medians. Because each window has its own Brown et al. threshold and direction, absolute percentages should be interpreted within window.

Near-eye fitted-curve dispersion and conditional Shapley allocation			
	Fitted-curve result (95% CI)	Conditional Shapley result (95% CI)	Share (95% CI)
Component summaries			
Shared local-clock curve	Not applicable	0.611 (0.550 to 0.657)	78.9% (73.6% to 82.6%)
Site pattern	0.100 (0.041 to 0.139)	0.016 (0.007 to 0.026)	2.0% (0.9% to 3.5%)
Participant pattern	0.180 (0.126 to 0.214)	0.100 (0.079 to 0.125)	12.9% (10.1% to 16.4%)
Participant-day shift	0.019 (0.010 to 0.024)	0.048 (0.035 to 0.064)	6.2% (4.4% to 8.6%)
Participant pattern + day shift	0.200 (0.140 to 0.230)	Not separately allocated	Not applicable
Participant-to-site comparisons			
Participant / site	1.797 (1.157 to 4.332)	6.41 (3.85 to 15.14)	Not applicable
(Participant + day) / site	1.991 (1.291 to 4.767)	9.49 (5.77 to 21.81)	90.5% (85.2% to 95.6%)

Full-model in-sample R² was 0.775 (0.739 to 0.804). Fitted-curve variation is in squared log₁₀(melEDI + 0.1 lx) prediction units; curve and Shapley ratios are unitless. Conditional Shapley values allocate in-sample R² with the shared local-clock curve as the mandatory baseline. Intervals are 95% percentile hierarchical cluster-bootstrap intervals from 2,000 replicates. Dispersion and Shapley credit are different estimands and must not be added or interpreted causally.

intervals from 2,000 replicates and are conditional on the fitted models. These quantities describe different estimands and should not be added or interpreted causally.

Supplementary Table S6. Chest fitted-curve dispersion and conditional Shapley allocation

Complementary chest fitted-curve dispersion and conditional Shapley allocation

	Fitted-curve result (95% CI)	Conditional Shapley result (95% CI)	Share (95% CI)
Component summaries			
Shared local-clock curve	Not applicable	0.583 (0.523 to 0.629)	79.1% (73.6% to 82.8%)
Site pattern	0.101 (0.049 to 0.116)	0.015 (0.008 to 0.024)	2.1% (1.1% to 3.3%)
Participant pattern	0.148 (0.100 to 0.199)	0.087 (0.071 to 0.110)	11.9% (9.5% to 15.3%)
Participant-day shift	0.035 (0.020 to 0.039)	0.051 (0.040 to 0.067)	7.0% (5.3% to 9.4%)
Participant pattern + day shift	0.183 (0.127 to 0.228)	Not separately allocated	Not applicable
Participant-to-site comparisons			
Participant / site	1.465 (0.973 to 3.179)	5.69 (3.72 to 10.68)	Not applicable
(Participant + day) / site	1.809 (1.219 to 3.736)	9.02 (6.01 to 17.03)	90.0% (85.7% to 94.5%)

Full-model in-sample R^2 was 0.737 (0.703 to 0.766). Fitted-curve variation is in squared $\log_{10}(\text{meEDI} + 0.1 \text{ lx})$ prediction units; curve and Shapley ratios are unitless. Conditional Shapley values allocate in-sample R^2 with the shared local-clock curve as the mandatory baseline. Intervals are 95% percentile hierarchical cluster-bootstrap intervals from 2,000 replicates. Dispersion and Shapley credit are different estimands and must not be added or interpreted causally.

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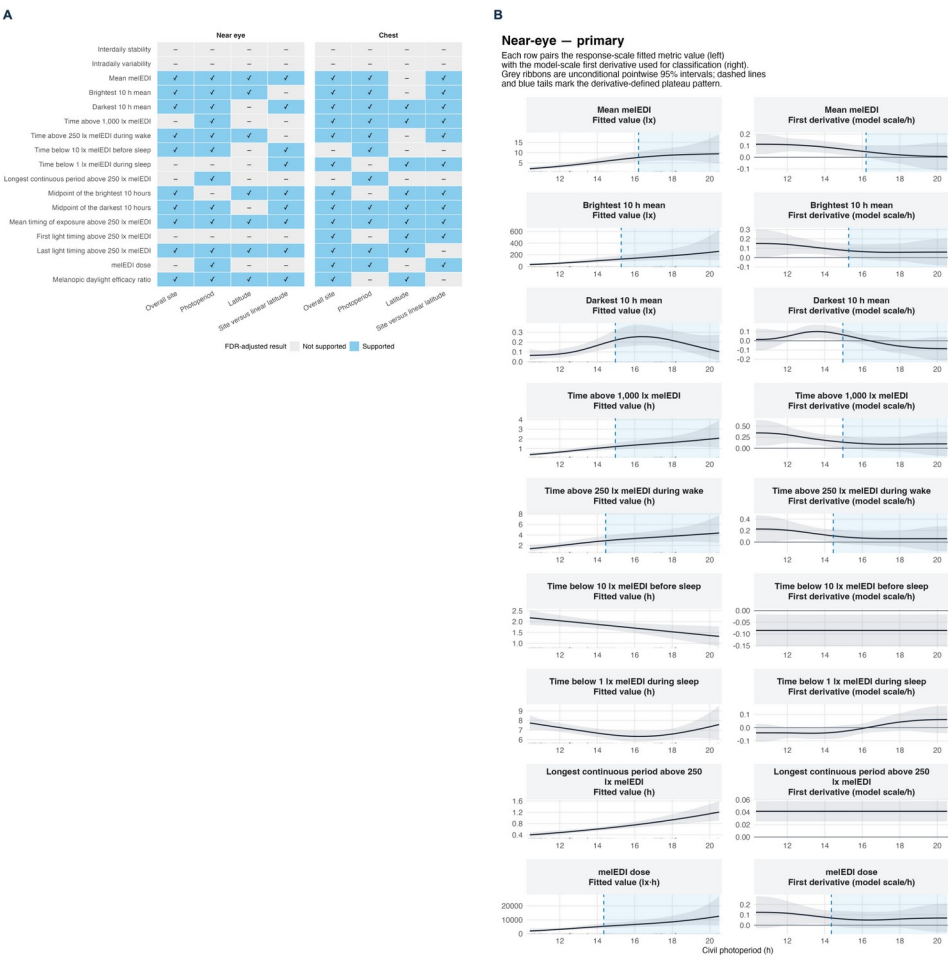
314 **Supplementary Table S6.** Complementary chest fitted-curve dispersion and conditional Shapley allocation of in-sample model fit. The quantities,
315 bootstrap intervals, and interpretation follow Supplementary Table S5; chest and near-eye measurements remain distinct sensor-position
316 estimands.

1317 **Geographic and civil-photoperiod context**

1318 **Supplementary Figure S7. Geographic support and nonlinear**

1319 **photoperiod classifications**

1320



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1322 **Supplementary Figure S7. Geographic and civil-photoperiod associations across**

1323 **near-eye personal light-exposure metrics. A, Evidence retained after FDR**

1324 **adjustment for overall site, civil photoperiod, latitude, and site-versus-linear-latitude**

1325 **adequacy. B, Nonlinear civil-photoperiod associations and their fitted slopes. A**

1326 **qualifying transition denotes an increase followed by a sustained near-flat tail under**

1327 **the specified slope rule. These descriptive associations do not identify a mechanism,**

1328 **and no physiological or environmental ceiling was identified.**

329 **Supplementary Table S7. Geographic and photoperiod associations**

		Photoperiod		Latitude		Site-versus-latitude FDR-adjusted p	Exact fitted sample
		Overall site FDR-adjusted p	Photoperiod association (95% CI)	Photoperiod FDR-adjusted p	Latitude association per 10° (95% CI)		
Dynamics							
Interdaily stability	0.228	×0.98 [0.94–1.02]	0.401	×1.01 [0.96–1.06]	0.903	0.175	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Intradaily variability	0.483	-0.02 [-0.06–0.02]	0.327	-0.04 [-0.08–0.01]	0.139	0.723	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Level							
Mean melEDI	<0.001	×1.18 [1.11–1.26]	<0.001	×1.15 [1.07–1.25]	0.002	0.032	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Brightest 10 h mean	0.011	×1.24 [1.13–1.36]	<0.001	×1.24 [1.11–1.39]	0.001	0.325	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Darkest 10 h mean	<0.001	×1.09 [1.03–1.14]	0.002	×1.04 [0.97–1.11]	0.400	<0.001	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$

Table continued

	Overall site FDR-adjusted p	Photoperiod		Latitude		Site-versus-latitude FDR-adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR-adjusted p	Latitude association per 10° (95% CI)	Latitude FDR-adjusted p		
Duration							
Time above 1,000 lx melEDI	0.071	×1.21 [1.13–1.29]	<0.001	×1.03 [0.94–1.12]	0.769	0.053	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Time above 250 lx melEDI during wake	0.007	×1.14 [1.07–1.20]	<0.001	×1.12 [1.04–1.20]	0.013	0.053	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 737$
Time below 10 lx melEDI before sleep	0.049	-0.13 [-0.20--0.07] h	<0.001	0.00 [-0.09–0.09] h	0.962	0.040	$n_{\text{participants}} = 139$; $n_{\text{participant-days}} = 655$
Time below 1 lx melEDI during sleep	0.058	×0.99 [0.97–1.01]	0.227	×1.00 [0.97–1.02]	0.903	0.043	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 778$
Longest continuous period above 250 lx melEDI	0.398	×1.12 [1.07–1.19]	<0.001	×1.06 [0.99–1.13]	0.139	0.626	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$

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Table continued

		Photoperiod		Latitude		Site-versus-latitude FDR-adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR-adjusted p	Latitude association per 10° (95% CI)	Latitude FDR-adjusted p		
Timing							
Midpoint of the brightest 10 hours	<0.001	0.05 [-0.05–0.15] h	0.305	0.19 [0.06–0.32] h	0.013	0.010	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 816
Midpoint of the darkest 10 hours	0.020	-0.15 [-0.25--0.05] h	0.004	0.12 [-0.00–0.24] h	0.114	0.043	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 816
Mean timing of exposure above 250 lx melEDI	<0.001	0.12 [0.03–0.21] h	0.013	0.33 [0.21–0.46] h	<0.001	<0.001	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 742
First light timing above 250 lx melEDI	0.112	-0.10 [-0.25–0.05] h	0.227	0.02 [-0.17–0.21] h	0.903	0.077	<i>n</i> _{participants} = 140; <i>n</i> _{participant-days} = 727
Last light timing above 250 lx melEDI	<0.001	0.32 [0.19–0.46] h	<0.001	0.46 [0.27–0.64] h	<0.001	<0.001	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 687
Exposure history							
melEDI dose	0.209	×1.28 [1.17–1.39]	<0.001	×1.02 [0.91–1.13]	0.903	0.156	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 761
Spectrum							
Melanopic daylight efficacy ratio	<0.001	0.02 [0.02–0.03]	<0.001	-0.01 [-0.02--0.00]	0.013	0.010	<i>n</i> _{participants} = 137; <i>n</i> _{participant-days} = 702
This summary combines the model-level decisions, photoperiod and latitude associations, 95% CIs, and exact fitted samples. Each displayed p-value is FDR-adjusted within its explicitly labelled complete 17-test family; bold values meet the FDR-adjusted p < 0.050 rule. Full raw p-values and test statistics remain in the detailed tables below. Exact fitted samples use italic n with participant and participant-day subscripts. For the 15 participant-day models, the participant-day count equals the number of fitted observations. The two participant-level dynamics models use one fitted observation per participant and retain the participant-day count as contributing repeated-day support. All models include nine sites.							

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333 **Supplementary Table S7.** Overall-site, civil-photoperiod and absolute-latitude associations for the 17 near-eye personal light-exposure metrics.
334 Each displayed p value is FDR-adjusted within its stated 17-test family. Site and latitude were evaluated in separate models because absolute
335 latitude is fixed within and confounded with study site.

	Qualifying transition	Qualifying transition bracket (h)	Endpoint derivative	Interpretation
Light level				
Mean melEDI	Yes	16.10–16.20	0.008 (95% CI -0.111 to 0.127)	Detected increase followed by a sustained zero-compatible tail
Brightest 10 h mean	Yes	15.17–15.27	0.059 (95% CI -0.083 to 0.201)	Detected increase followed by a sustained zero-compatible tail
Darkest 10 h mean	Yes	14.86–14.96	-0.086 (95% CI -0.216 to 0.044)	Detected increase followed by a sustained zero-compatible tail
Duration and continuous period				
Time above 1,000 lx melEDI	Yes	14.86–14.96	0.098 (95% CI -0.176 to 0.371)	Detected increase followed by a sustained zero-compatible tail
Time above 250 lx melEDI during wake	Yes	14.35–14.45	0.060 (95% CI -0.159 to 0.279)	Detected increase followed by a sustained zero-compatible tail
Time below 10 lx melEDI before sleep	No	—	-0.085 (95% CI -0.154 to -0.016)	No preceding detected increase
Time below 1 lx melEDI during sleep	No	—	0.062 (95% CI -0.042 to 0.165)	No preceding detected increase
Longest continuous period above 250 lx melEDI	No	—	0.041 (95% CI 0.025 to 0.058)	Increase remained detected at the recorded maximum
Exposure history				
melEDI dose	Yes	14.24–14.35	0.070 (95% CI -0.069 to 0.209)	Detected increase followed by a sustained zero-compatible tail
A 'Yes' is the formal derivative-defined classification: the immediately preceding point has an interval wholly above zero, the next contains zero, and every later interval through the recorded maximum remains zero-compatible. Endpoint derivatives and 95% confidence intervals are on each model's linear-predictor scale, in model-scale units per hour of civil photoperiod.				

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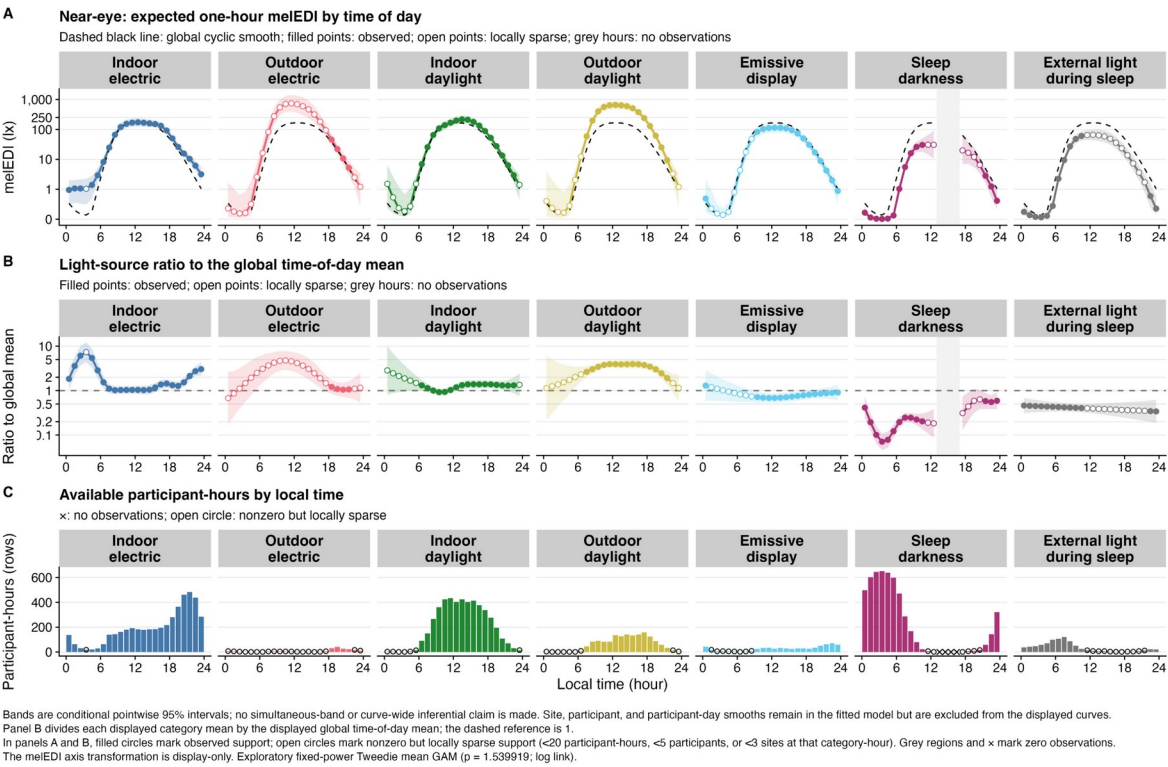
338 **Supplementary Table S8.** Derivative-based classifications for nine near-eye personal light-exposure metrics across the observed civil-
339 photoperiod range. A qualifying transition required the preceding fitted-slope interval to be wholly above zero, the next interval to include zero and
340 every later interval through the recorded maximum to remain zero-compatible. These classifications are descriptive and do not identify a
341 physiological or environmental ceiling.

1342 Immediate environments and routines

1343 **Supplementary Figure S8. Light-source category and local-clock**

1344 **pattern of near-eye melanopic EDI**

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D Near-eye light-source-by-site interaction model
Site-average estimates and site-specific deviation ratios; bold site cells pass the complete near-eye FDR adjustment

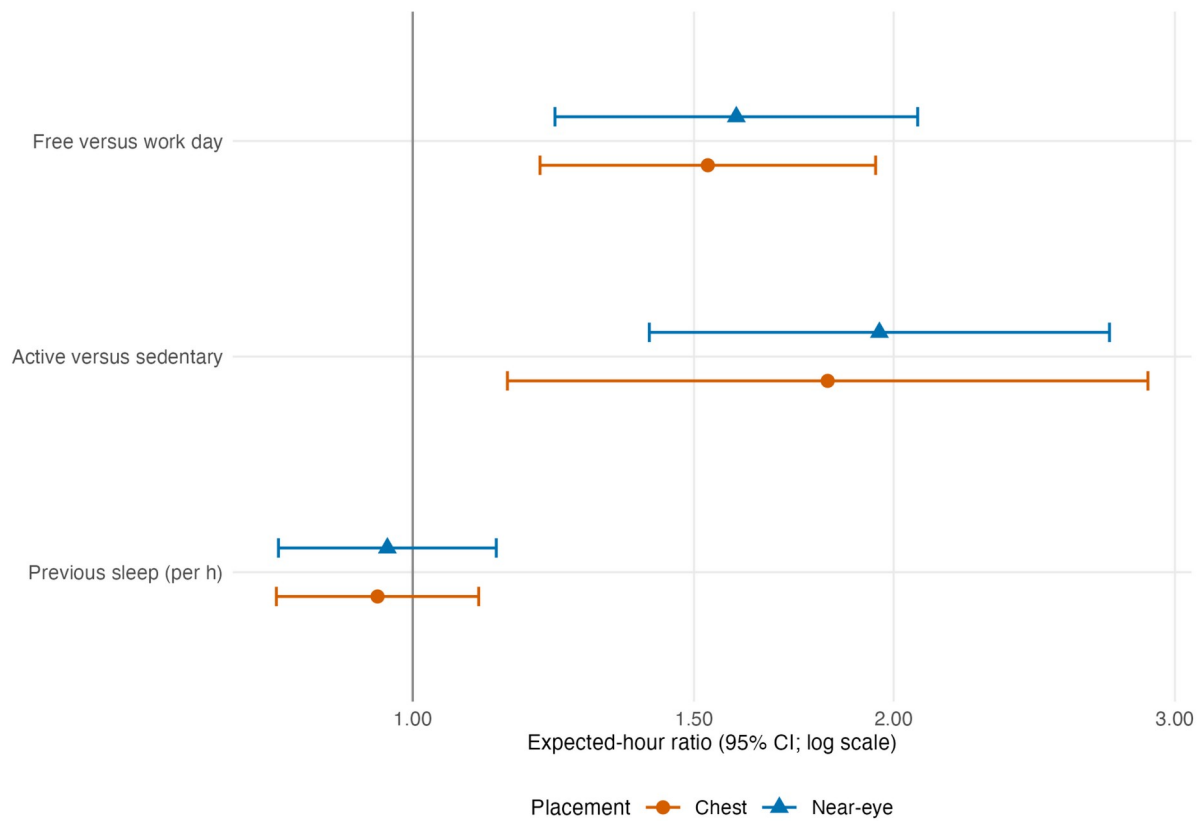
	Indoor electric	Outdoor electric	Indoor daylight	Outdoor daylight	Emissive display	Sleep darkness	External light during sleep
	P 138 · D 695 · H 4,629	P 54 · D 118 · H 203	P 138 · D 692 · H 4,804	P 135 · D 512 · H 1,557	P 88 · D 235 · H 664	P 130 · D 712 · H 5,225	P 66 · D 233 · H 853
Site-average estimate	85.1 lx 95% CI 71.6–101.1 1.000× reference	9.4 lx 95% CI 6.9–12.8 0.110× indoor 95% CI 0.078–0.156	198.8 lx 95% CI 169.2–233.4 2.335× indoor 95% CI 1.877–2.905	959.6 lx 95% CI 786.8–1170.2 11.274× indoor 95% CI 8.709–14.595	23.8 lx 95% CI 17.4–32.5 0.280× indoor 95% CI 0.197–0.398	2.2 lx 95% CI 1.5–3.3 0.026× indoor 95% CI 0.018–0.039	11.2 lx 95% CI 8.1–15.5 0.131× indoor 95% CI 0.091–0.189
Borås (SE)	1.43× (1.02–1.99) FDR p 0.083	Not estimable	1.61× (1.12–2.32) FDR p 0.029	3.07× (1.92–4.90) FDR p <0.001	0.31× (0.16–0.57) FDR p 0.002	1.52× (0.39–5.90) FDR p 0.704	0.70× (0.21–2.36) FDR p 0.707
Delft (NL)	1.83× (1.32–2.54) FDR p 0.002	Not estimable	1.46× (0.93–2.29) FDR p 0.190	1.58× (0.89–2.80) FDR p 0.214	7.17× (4.00–12.86) FDR p <0.001	0.94× (0.49–1.80) FDR p 0.894	Not estimable
Dortmund (DE)	1.26× (0.69–2.29) FDR p 0.595	Not estimable	1.62× (0.83–3.17) FDR p 0.263	1.48× (0.91–2.40) FDR p 0.210	2.86× (1.66–4.93) FDR p 0.002	0.89× (0.45–1.75) FDR p 0.801	9.22× (2.50–34.03) FDR p 0.004
Tübingen (DE)	0.56× (0.38–0.82) FDR p 0.009	20.42× (3.95–105.59) FDR p 0.002	1.04× (0.72–1.50) FDR p 0.866	0.89× (0.51–1.56) FDR p 0.776	0.25× (0.14–0.44) FDR p <0.001	0.78× (0.32–1.92) FDR p 0.710	1.59× (0.71–3.59) FDR p 0.406
Munich (DE)	1.70× (0.63–4.57) FDR p 0.425	Not estimable	1.00× (0.65–1.55) FDR p 0.983	0.94× (0.55–1.61) FDR p 0.866	Not estimable	6.60× (2.11–20.69) FDR p 0.005	11.65× (7.19–18.85) FDR p <0.001
Madrid (ES)	1.48× (1.15–1.90) FDR p 0.007	2.68× (1.56–4.61) FDR p 0.002	0.83× (0.57–1.19) FDR p 0.436	0.65× (0.33–1.26) FDR p 0.332	3.68× (2.24–6.03) FDR p <0.001	1.59× (0.30–8.32) FDR p 0.710	Not estimable
Izmir (TR)	0.69× (0.52–0.92) FDR p 0.029	1.85× (0.84–4.07) FDR p 0.222	0.86× (0.65–1.13) FDR p 0.422	1.12× (0.65–1.94) FDR p 0.776	1.32× (0.70–2.47) FDR p 0.540	0.97× (0.43–2.18) FDR p 0.952	1.59× (0.92–2.76) FDR p 0.190
San José (CR)	0.54× (0.38–0.77) FDR p 0.003	Not estimable	0.72× (0.49–1.05) FDR p 0.189	0.46× (0.21–1.01) FDR p 0.115	1.32× (0.65–2.66) FDR p 0.595	0.60× (0.25–1.42) FDR p 0.390	Not estimable
Kumasi (GH)	0.58× (0.38–0.88) FDR p 0.029	6.24× (2.28–17.06) FDR p 0.002	0.49× (0.26–0.92) FDR p 0.065	0.50× (0.36–0.68) FDR p <0.001	1.27× (0.40–4.07) FDR p 0.776	0.17× (0.06–0.49) FDR p 0.004	Not estimable

Site-average estimates and comparisons use the light-source-by-site interaction model.
Site rows are deviation ratios around each category estimate; parentheses show 95% confidence intervals.
Bold site ratios and FDR p-values pass the complete 53-member near-eye family; unsupported cells are not estimable.
Near-eye frame: 17,935 participant-hours; 140 participants; 801 participant-days; nine sites.
Each accepted participant-hour has one primary light-source category; 1/k multi-label weighting does not apply.
Panels A–C exclude site, participant, and participant-day smooths from displayed curves; panel D uses stored interaction-model estimands.
P, participants; D, participant-days; H, participant-hours. Country-coded labels and row order provide non-colour site cues.

Supplementary Figure S8. Near-eye light exposure by reported light source across time of day and study site. A–C, Exploratory time-of-day analysis. Panel A shows expected one-hour melanopic EDI with clock-specific 95% confidence intervals; the dashed line is the site-average local-clock smooth. Panel B shows each light-source curve relative to that smooth, with 1 as the reference. Panel C shows available participant-hours; open circles indicate locally sparse support and grey gaps indicate no observations. D, Site-average category estimates and site-specific deviation ratios from the light-source-by-site interaction model, with participant-cluster-robust 95% confidence intervals, FDR-adjusted p values, and exact support. The analysis includes 17,935 participant-hours from 140 participants, covering 801 participant-days and nine sites. Panels A–C and panel D answer different questions and are not numerically interchangeable. Estimates are observational. During reported sleep, measurements describe the bedside sleep environment.

Hourly routine analyses

Supplementary Figure S9. Paired sensor-position hourly associations

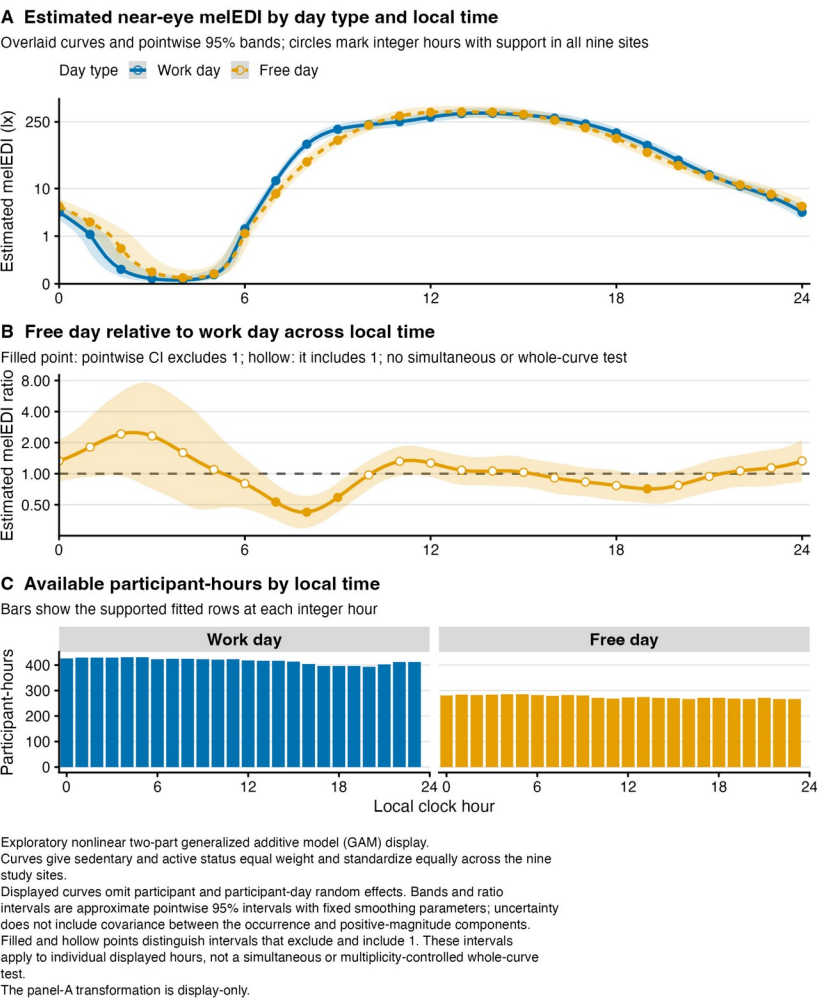


1366

1367 Supplementary Figure S9. Near-eye and complementary chest associations
 1368 estimated separately for the same participants and participant-days. The sets of
 1369 supported participant-hours differ between sensor positions, so the display is not an
 1370 observation-level identity comparison and does not test equivalence.

1371 **Supplementary Figure S10. Free-versus-Work hourly timing**

1372

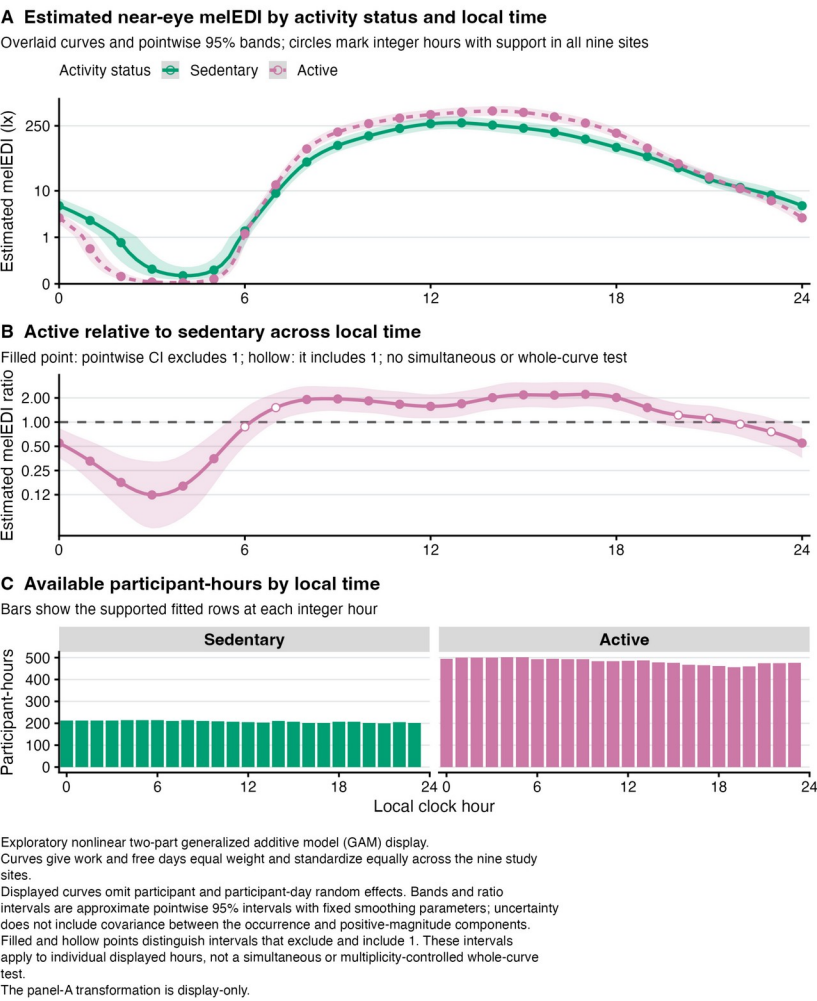


1373

1374 Supplementary Figure S10. Exploratory Free-versus-Work hourly fitted exposure and
1375 response-scale ratio curves with 95% confidence intervals and participant-hour
1376 support.

1377 **Supplementary Figure S11. Active-versus-Sedentary hourly timing**

1378

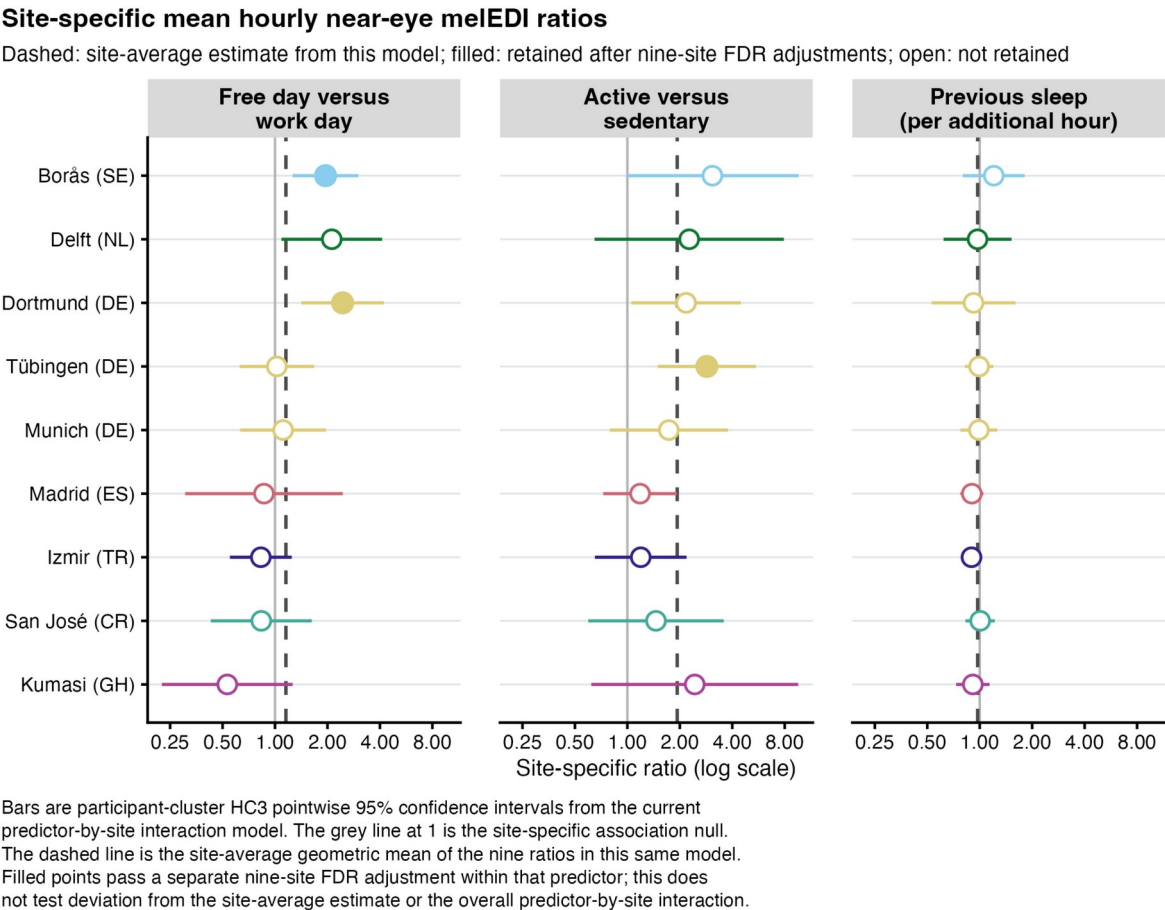


1379

1380 Supplementary Figure S11. Exploratory Active-versus-Sedentary hourly fitted
1381 exposure and response-scale ratio curves with 95% confidence intervals and
1382 participant-hour support.

1383 **Supplementary Figure S12. Site-specific hourly routine**
1384 **associations**

1385



1386

1387 Supplementary Figure S12. Site-specific estimates from the predictor-by-site
1388 interaction model. Filled and hollow points distinguish associations that did and did
1389 not meet the 0.050 criterion after a separate nine-site FDR adjustment within each
1390 predictor. The 95% confidence intervals are unadjusted; dashed lines show site-
1391 average estimates calculated with equal weight for each of the nine sites. Sites are
1392 components of one pooled model, not independent replications or causal effects of
1393 location.

394 **Supplementary Table S9. Participant-hour routine associations**

Predictor	Ratio (95% CI)	FDR-adjusted p
Free day versus work day	1.45 × (1.14–1.85)	0.004
Active versus sedentary	2.06 × (1.54–2.76)	<0.001
Previous sleep duration (per hour)	0.98 × (0.85–1.12)	0.745
Directions are free versus work day, active versus sedentary day status, and the ratio per one additional hour of previous sleep. Participant-cluster-robust 95% CIs use HC3 with 137 participants. The three p-values form one FDR-adjusted family; the null ratio is 1.		

395
396 **Supplementary Table S9.** Common-association estimates from the participant-hour analysis of Free versus Work day, Active versus Sedentary
397 daily activity status and each additional hour of previous-night sleep duration. Ratios and participant-cluster-robust 95% confidence intervals come
398 from the fixed-site model; the three *p* values form one FDR-adjusted family. Predictor-by-site differences are shown in Supplementary Figure S12.

Person-level evidence synthesis						
	Outcome or analysis scale	Result	FDR-correction set	FDR decision	Exact sample	Interpretation and qualification
Light-exposure behaviour and awareness factors	68 factor-by-metric cells across four participant-level factors	No cell retained FDR support; four sleep-environment cells were non-estimable	68 near-eye factor-by-metric tests	None retained	Metric-specific near-eye samples in Supplementary Table S11	Broad inconclusive result; non-retention is not evidence of no association
Visual light sensitivity	Nine near-eye metrics per one VLSQ-8 SD	Largest displayed ratio shift was melanopic EDI dose, $\times 0.846$ (0.716 to 0.999)	Nine near-eye metric tests	None retained; adjusted $p = 0.160$ for this row	141 participants, 761 participant-days, nine sites for the displayed row	Broad inconclusive result
Corrected midsleep on Free days	Timing outcomes per one-hour later MSFsc	M10 midpoint +0.205 h (0.045 to 0.366); L10 midpoint +0.276 h (0.118 to 0.435); first light above 250 lx +0.381 h (0.146 to 0.616)	Five MCTQ timing outcomes, separate from MEQ	These three rows retained	139 to 140 participants; 722 to 810 participant-days; nine sites	Selective timing associations
Morningness-Eveningness Questionnaire	Timing outcomes per 10 points greater morning preference	M10 midpoint -0.278 h (-0.451 to -0.105); L10 midpoint -0.314 h (-0.487 to -0.142); first light above 250 lx -0.444 h (-0.699 to -0.189)	Five MEQ timing outcomes, separate from MCTQ	These three rows retained	140 to 141 participants; 727 to 816 participant-days; nine sites	Selective timing associations

Table continued

Person-level evidence synthesis						
	Outcome or analysis scale	Result	FDR-correction set	FDR decision	Exact sample	Interpretation and qualification
Age	Near-eye daily metrics per 10 years	Brightest 10 h geometric mean $\times 1.31$ (1.09 to 1.58); time above 1,000 lx $\times 1.27$ (1.11 to 1.45); melanopic EDI dose $\times 1.31$ (1.09 to 1.56)	Near-eye age-metric tests	Three rows retained	141 participants; 761 to 816 participant-days; nine sites	Selective age associations; complementary chest results remain separate
Metric-level biological sex	Near-eye daily metrics, Female minus Male	No near-eye metric-level result retained FDR support	Near-eye biological-sex metric tests	None retained	Metric-specific samples in Supplementary Table S14	Biological sex and gender were recorded separately; gender was not analysed
Biological-sex-specific daily curve	Complete 24-hour curve and activity-complete sensitivity	The all-available near-eye complete-curve test was supported, but separation attenuated and was not retained in the activity-complete near-eye analysis	Global complete-curve tests and separately prespecified sensitivity decisions	Analysis-specific global decisions	All-available near eye: 141 participants, 816 participant-days, 37,756 observations, nine sites	Attenuation-first interpretation; confidence intervals apply at the displayed clock times

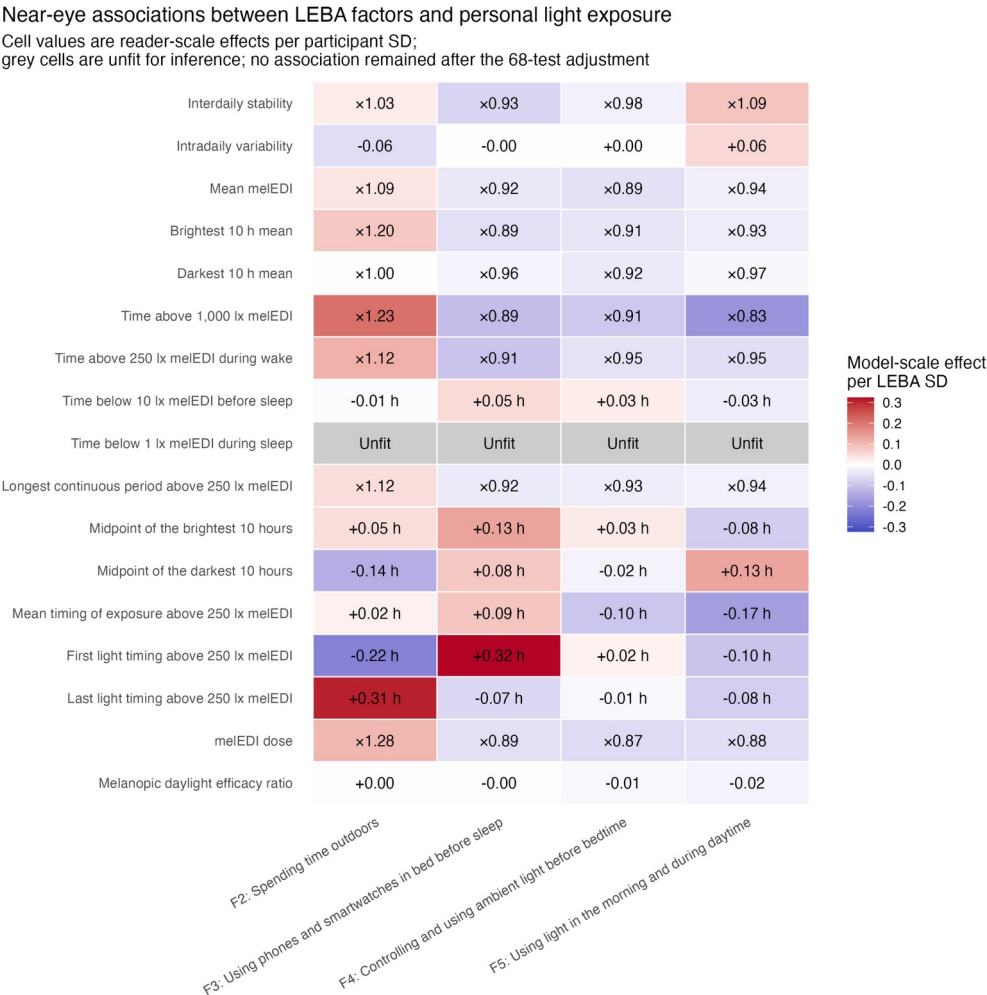
Results that did not meet the applicable FDR criterion are inconclusive rather than evidence of no association. The two chronotype instruments were analysed separately. Biological sex and gender were recorded separately; gender was not analysed.

Supplementary Table S10. Summary of person-level results in the near-eye analyses. Each row identifies the analysis-specific FDR-correction set or decision structure applied to that result. Results that did not meet the applicable FDR criterion are inconclusive rather than evidence of no association. Four light-behaviour sleep-environment cells were non-estimable. The two chronotype instruments were analysed separately.

406 Biological sex and gender were recorded separately; gender was not analysed. The complete available-data curve difference was attenuated and
407 not retained in the activity-complete near-eye analysis.

1408 **Supplementary Figure S13 and Table S11. Light-exposure**
1409 **behaviour and awareness**

1410



1411
1412 **Supplementary Figure S13. Near-eye associations per one participant-level standard**
1413 **deviation of each Light Exposure Behaviour Assessment factor. None of the 68 tests**
1414 **retained FDR-adjusted support. Four sleep-environment cells were not estimable**
1415 **and are labelled Unfit with estimates suppressed. Lack of adjusted support is**
1416 **inconclusive rather than proof of no association.**

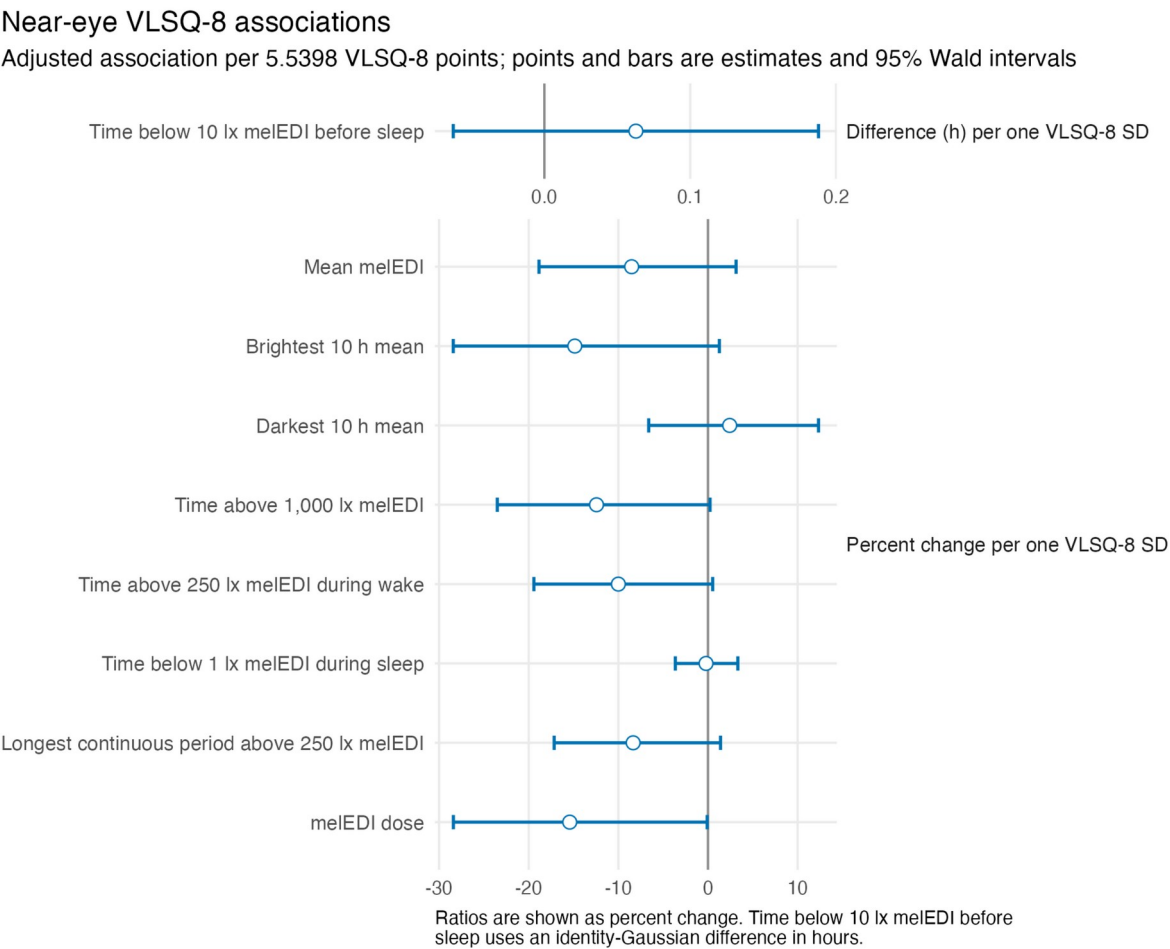
	F2	F3	F4	F5
Interdaily stability	OR 1.028 (0.956–1.106) FDR-adjusted p = 0.658	OR 0.931 (0.868–0.999) FDR-adjusted p = 0.260	OR 0.975 (0.906–1.050) FDR-adjusted p = 0.679	OR 1.089 (1.011–1.174) FDR-adjusted p = 0.202
Intradaily variability	-0.057 (-0.124–+0.009) FDR-adjusted p = 0.351	-0.001 (-0.068–+0.065) FDR-adjusted p = 0.978	+0.001 (-0.068–+0.069) FDR-adjusted p = 0.978	+0.057 (-0.013–+0.127) FDR-adjusted p = 0.362
Mean melEDI	×1.092 (0.965–1.235) FDR-adjusted p = 0.379	×0.917 (0.813–1.034) FDR-adjusted p = 0.379	×0.891 (0.787–1.009) FDR-adjusted p = 0.313	×0.941 (0.827–1.070) FDR-adjusted p = 0.585
Brightest 10 h mean	×1.204 (1.008–1.439) FDR-adjusted p = 0.260	×0.888 (0.745–1.058) FDR-adjusted p = 0.384	×0.906 (0.756–1.087) FDR-adjusted p = 0.545	×0.931 (0.772–1.123) FDR-adjusted p = 0.658
Darkest 10 h mean	×0.995 (0.905–1.094) FDR-adjusted p = 0.958	×0.956 (0.871–1.049) FDR-adjusted p = 0.577	×0.921 (0.838–1.013) FDR-adjusted p = 0.351	×0.970 (0.879–1.070) FDR-adjusted p = 0.716
Time above 1,000 lx melEDI	×1.234 (1.084–1.405) FDR-adjusted p = 0.124	×0.895 (0.785–1.020) FDR-adjusted p = 0.362	×0.911 (0.797–1.043) FDR-adjusted p = 0.390	×0.831 (0.725–0.953) FDR-adjusted p = 0.169
Time above 250 lx melEDI during wake	×1.122 (1.006–1.252) FDR-adjusted p = 0.260	×0.907 (0.814–1.010) FDR-adjusted p = 0.351	×0.953 (0.852–1.065) FDR-adjusted p = 0.614	×0.946 (0.843–1.061) FDR-adjusted p = 0.586
Time below 10 lx melEDI before sleep	-0.005 h (-0.135–+0.124 h) FDR-adjusted p = 0.958	+0.052 h (-0.073–+0.177 h) FDR-adjusted p = 0.614	+0.028 h (-0.104–+0.159 h) FDR-adjusted p = 0.781	-0.031 h (-0.167–+0.105 h) FDR-adjusted p = 0.781
Time below 1 lx melEDI during sleep	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display
Longest continuous period above 250 lx melEDI	×1.117 (1.007–1.238) FDR-adjusted p = 0.260	×0.924 (0.835–1.023) FDR-adjusted p = 0.375	×0.933 (0.840–1.037) FDR-adjusted p = 0.393	×0.945 (0.847–1.053) FDR-adjusted p = 0.557

Associations are reported per one participant-level SD of the LEBA score with 95% CIs. FDR-adjusted p is bold only at alpha = 0.050 across all 68 tests. Sleep-environment cells are unfit for inference and suppressed.

	F2	F3	F4	F5
Midpoint of the brightest 10 hours	+0.049 h (-0.132–+0.229 h) FDR-adjusted p = 0.747	+0.133 h (-0.042–+0.309 h) FDR-adjusted p = 0.375	+0.030 h (-0.153–+0.213 h) FDR-adjusted p = 0.868	-0.083 h (-0.270–+0.105 h) FDR-adjusted p = 0.601
Midpoint of the darkest 10 hours	-0.136 h (-0.317–+0.044 h) FDR-adjusted p = 0.375	+0.080 h (-0.097–+0.258 h) FDR-adjusted p = 0.594	-0.020 h (-0.204–+0.164 h) FDR-adjusted p = 0.935	+0.132 h (-0.056–+0.320 h) FDR-adjusted p = 0.379
Mean timing of exposure above 250 lx melEDI	+0.020 h (-0.152–+0.192 h) FDR-adjusted p = 0.935	+0.086 h (-0.081–+0.253 h) FDR-adjusted p = 0.558	-0.097 h (-0.267–+0.074 h) FDR-adjusted p = 0.510	-0.166 h (-0.339–+0.008 h) FDR-adjusted p = 0.301
First light timing above 250 lx melEDI	-0.219 h (-0.489–+0.052 h) FDR-adjusted p = 0.362	+0.324 h (+0.062–+0.585 h) FDR-adjusted p = 0.179	+0.019 h (-0.252–+0.291 h) FDR-adjusted p = 0.958	-0.097 h (-0.380–+0.185 h) FDR-adjusted p = 0.679
Last light timing above 250 lx melEDI	+0.308 h (+0.043–+0.573 h) FDR-adjusted p = 0.202	-0.066 h (-0.328–+0.196 h) FDR-adjusted p = 0.760	-0.011 h (-0.280–+0.259 h) FDR-adjusted p = 0.958	-0.081 h (-0.361–+0.200 h) FDR-adjusted p = 0.743
melEDI dose	×1.278 (1.079–1.514) FDR-adjusted p = 0.124	×0.885 (0.748–1.048) FDR-adjusted p = 0.379	×0.872 (0.732–1.038) FDR-adjusted p = 0.375	×0.881 (0.736–1.054) FDR-adjusted p = 0.379
Melanopic daylight efficacy ratio	+0.004 (-0.011–+0.019) FDR-adjusted p = 0.747	-0.004 (-0.019–+0.011) FDR-adjusted p = 0.747	-0.010 (-0.026–+0.005) FDR-adjusted p = 0.380	-0.020 (-0.035–-0.004) FDR-adjusted p = 0.169
Associations are reported per one participant-level SD of the LEBA score with 95% CIs. FDR-adjusted p is bold only at alpha = 0.050 across all 68 tests. Sleep-environment cells are unfit for inference and suppressed.				

1422 **Supplementary Figure S14 and Table S12. Visual light sensitivity**

1423



1424

1425 Supplementary Figure S14. Near-eye site-average associations between the Visual
1426 Light Sensitivity Questionnaire-8 and nine personal light-exposure metrics.

1427 Associations compare scores separated by one participant-level standard deviation.

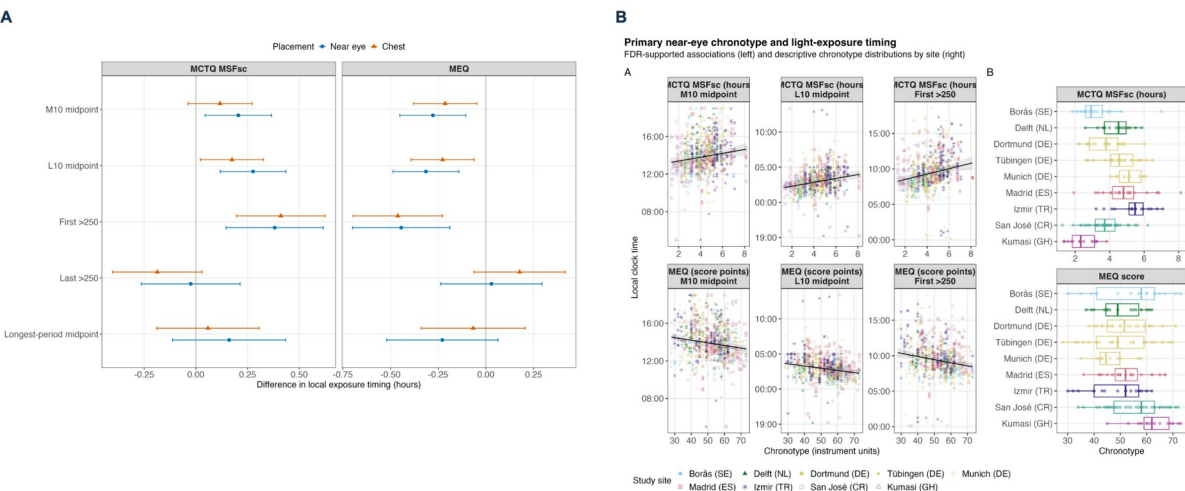
1428 Horizontal bars are 95% Wald confidence intervals. None of the nine associations

1429 retained FDR-adjusted support.

Metric	Scale	Association per VLSQ-8 SD (95% CI)	FDR-adjusted p	Participants / participant-days / sites
Mean melEDI	Ratio (percentage change)	×0.915 (0.811–1.031)	0.221	141 / 816 / 9
Brightest 10 h mean	Ratio (percentage change)	×0.851 (0.716–1.013)	0.160	141 / 816 / 9
Darkest 10 h mean	Ratio (percentage change)	×1.024 (0.934–1.123)	0.689	141 / 816 / 9
Time above 1,000 lx melEDI	Ratio (percentage change)	×0.876 (0.765–1.002)	0.160	141 / 816 / 9
Time above 250 lx melEDI during wake	Ratio (percentage change)	×0.900 (0.806–1.005)	0.160	141 / 737 / 9
Time below 10 lx melEDI before sleep	Difference in hours	+0.063 h (-0.062–+0.188 h)	0.422	139 / 655 / 9
Time below 1 lx melEDI during sleep	Ratio (percentage change)	×0.998 (0.964–1.033)	0.903	141 / 778 / 9
Longest continuous period above 250 lx melEDI	Ratio (percentage change)	×0.917 (0.828–1.014)	0.167	141 / 816 / 9
melEDI dose	Ratio (percentage change)	×0.846 (0.716–0.999)	0.160	141 / 761 / 9
Associations compare scores separated by one participant-level SD (5.540 VLSQ-8 points). The hour-scale row is an absolute difference; ratio rows can be read as percentage changes. Intervals are two-sided 95% Wald CIs. FDR adjustment is across all nine metrics; adjusted p-values would be bold at 0.050.				

1432 **Supplementary Figure S15 and Table S13. Chronotype and timing**

1433



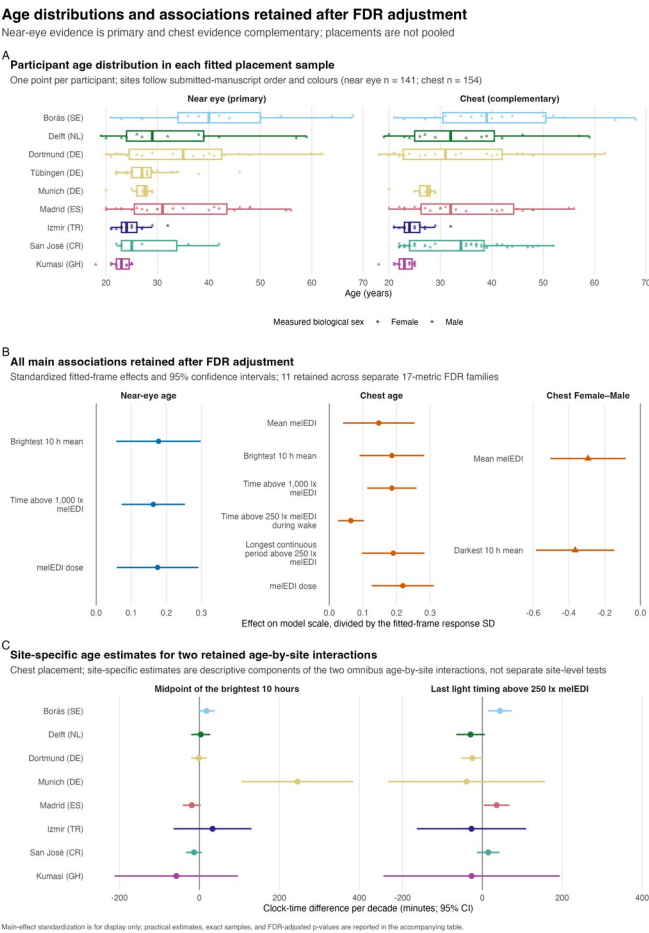
1434

1435 Supplementary Figure S15. Chronotype and timing of personal light exposure. A,
1436 Study-site-adjusted associations of sleep-timing-based corrected midsleep on Free
1437 days and questionnaire-based morningness-eveningness with the analysed timing
1438 metrics. Points are estimates and bars are 95% confidence intervals; the two
1439 chronotype instruments remain separate. B, Observed participant-day timing values
1440 across chronotype and country-coded study sites for the samples used in the
1441 corresponding models. Each point represents one participant-day in a fitted sample.
1442 The observed panel is descriptive and does not replace the adjusted models or
1443 estimate independent site effects. Near-eye estimates are primary; chest estimates,
1444 where shown, are complementary non-ocular evidence.

Metric	Instrument	Signed association, h (95% CI)	FDR-adjusted p	Fitted sample: participants / days / observations / hours / sites
Midpoint of the brightest 10 hours	MCTQ MSFsc	+0.205 (+0.045 to +0.366)	0.017	140 / 810 / 810 / 18711.7 h / 9
Midpoint of the brightest 10 hours	MEQ	-0.278 (-0.451 to -0.105)	0.002	141 / 816 / 816 / 18851.0 h / 9
Midpoint of the darkest 10 hours	MCTQ MSFsc	+0.276 (+0.118 to +0.435)	0.003	140 / 810 / 810 / 18711.7 h / 9
Midpoint of the darkest 10 hours	MEQ	-0.314 (-0.487 to -0.142)	0.001	141 / 816 / 816 / 18851.0 h / 9
First light timing above 250 lx meLEDI	MCTQ MSFsc	+0.381 (+0.146 to +0.616)	0.003	139 / 722 / 722 / 16716.3 h / 9
First light timing above 250 lx meLEDI	MEQ	-0.444 (-0.699 to -0.189)	0.001	140 / 727 / 727 / 16832.5 h / 9
Last light timing above 250 lx meLEDI	MCTQ MSFsc	-0.025 (-0.264 to +0.214)	0.811	140 / 683 / 683 / 15900.8 h / 9
Last light timing above 250 lx meLEDI	MEQ	+0.029 (-0.236 to +0.294)	0.814	141 / 687 / 687 / 15995.0 h / 9
Midpoint of the longest continuous period above 250 lx meLEDI	MCTQ MSFsc	+0.161 (-0.113 to +0.434)	0.284	131 / 478 / 478 / 11325.5 h / 9
Midpoint of the longest continuous period above 250 lx meLEDI	MEQ	-0.228 (-0.521 to +0.064)	0.135	132 / 482 / 482 / 11419.8 h / 9
MCTQ associations are per one-hour later MSFsc; MEQ associations are per 10 points greater morning preference. Positive associations indicate later timing and negative associations earlier. Intervals are 95% Wald CIs. FDR-adjusted p-values are bold when FDR-adjusted p < 0.050 within the instrument-specific five-outcome family.				

1447 **Supplementary Figure S16 and Table S14. Age and biological sex**

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1449

1450 Supplementary Figure S16. Participant age distributions, the 11 main associations

1451 retained after FDR adjustment, and descriptive site-specific estimates for two

1452 retained age-by-site interactions in the complementary chest analysis. The site-

1453 specific interaction estimates are descriptive model contrasts, not independent site

1454 tests. Biological sex, coded Female or Male, was analysed; gender was recorded

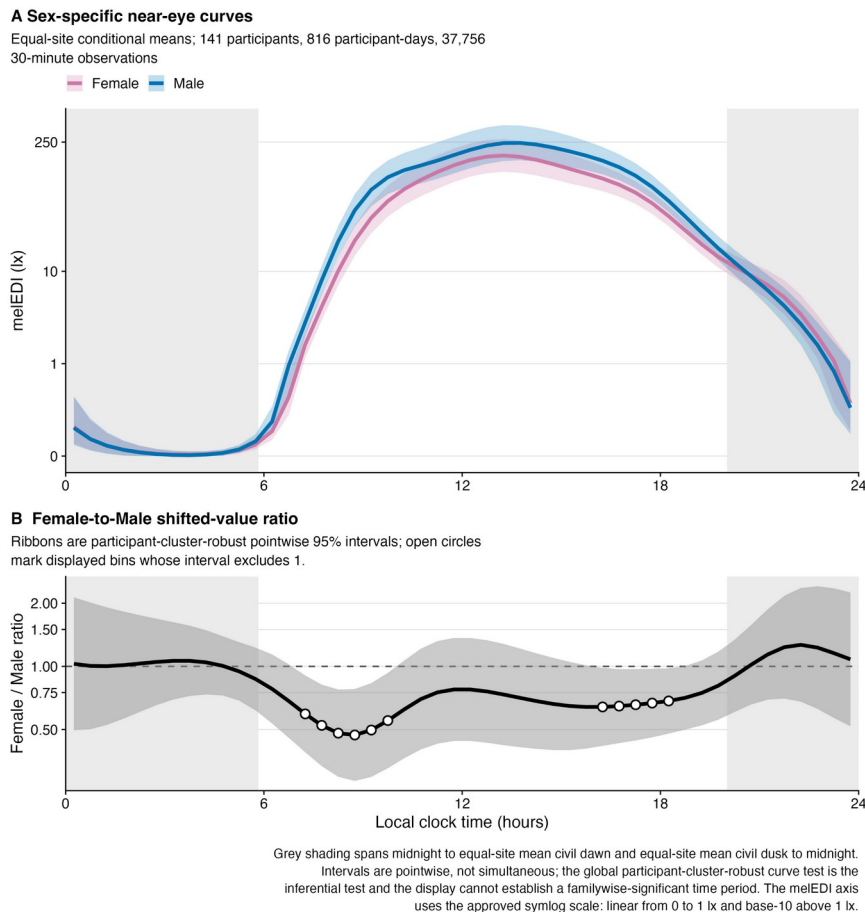
1455 separately and was not analysed.

Placement and role	Predictor	Metric	Practical association (95% CI)	FDR-adjusted p	Sample
Near eye (primary)	Age, per 10 years	Brightest 10 h mean	1.31× (1.09–1.58)	0.018	141 participants; 816 participant-days/observations
Near eye (primary)	Age, per 10 years	Time above 1,000 lx melEDI	1.27× (1.11–1.45)	0.009	141 participants; 816 participant-days/observations
Near eye (primary)	Age, per 10 years	melEDI dose	1.31× (1.09–1.56)	0.018	141 participants; 761 participant-days/observations
Chest (complementary)	Age, per 10 years	Mean melEDI	1.16× (1.04–1.30)	0.017	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Brightest 10 h mean	1.35× (1.15–1.57)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Time above 1,000 lx melEDI	1.31× (1.18–1.46)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Time above 250 lx melEDI during wake	1.16× (1.06–1.26)	0.004	154 participants; 818 participant-days/observations
Chest (complementary)	Age, per 10 years	Longest continuous period above 250 lx melEDI	1.18× (1.09–1.27)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	melEDI dose	1.40× (1.21–1.61)	<0.001	154 participants; 851 participant-days/observations
Chest (complementary)	Measured biological sex, Female minus Male	Mean melEDI	0.74× (0.60–0.92)	0.048	154 participants; 902 participant-days/observations
Chest (complementary)	Measured biological sex, Female minus Male	Darkest 10 h mean	0.76× (0.65–0.90)	0.016	154 participants; 902 participant-days/observations

Associations are site-adjusted estimates per 10-year age increase or Female minus Male. Ratios are back-transformed to the practical scale. Each FDR-adjusted p-value belongs to its sensor-position- and predictor-specific complete 17-test family; bold values meet the FDR-adjusted p < 0.050 rule. Samples give the exact fitted participants and participant-day observations for these retained daily-response results.

1458 **Supplementary Figure S17 and Table S15. Biological-sex-specific**
1459 **daily curves**

1460



1461
1462 Supplementary Figure S17. Near-eye biological-sex-specific fitted melanopic
1463 equivalent daylight illuminance curves and Female-to-Male shifted-value ratio.
1464 Ribbons and contrast intervals are participant-cluster-robust 95% confidence
1465 intervals at the displayed clock times. Grey shading gives site-average civil-night
1466 context with each site weighted equally and is not a model covariate. Biological sex,
1467 coded Female or Male, was analysed; gender was recorded separately and was not
1468 analysed.

	Role	Test	F statistic	Numerator df	Denominator df	Raw p	FDR-adjusted p	Support
Near eye	Primary	Complete Female-minus-Male 24-hour curve	20.946	9.692	131	0.028	0.028	Supported
Chest	Complementary	Complete Female-minus-Male 24-hour curve	4.876	1.003	152	0.029	0.029	Supported

Primary near-eye fitted sample: 141 participants, 816 participant-days, 37,756 30-minute observations, and 9 sites. Complementary chest fitted sample: 154 participants, 902 participant-days, 41,842 30-minute observations, and 8 sites.

Each row is one joint complete-curve test. Raw p is bold at raw $p < 0.050$; FDR-adjusted p is bold independently at adjusted $p < 0.050$. Each labelled global family contains one test, so the stored raw and adjusted values coincide.

1471 **Measurement position, robustness, and reproducibility**

1472 Near-eye and complementary chest measurements remain separate sensor-position
1473 estimands. Common-sample analyses use the same participants and participant-
1474 days at both positions and fit the positions separately. They do not pool positions,
1475 test equivalence, or establish a universal correction. Detailed model checks,
1476 sensitivities, preregistration deviations, exact formulas, package versions, and
1477 source documentation are provided in the analysis reports and reproducibility
1478 package.